HRMS Workforce Management & Payroll System

Software Requirement Specification (SRS)

Version 1.0

Chapter 1

System Vision

1.1 Objective

The purpose of this project is to build a modern Enterprise Workforce Management and Payroll platform capable of replacing legacy Payroll, Timesheet and Workforce systems used by large organizations.

The application is intended for organizations operating in Construction, Oil & Gas, Facilities Management, Manufacturing, Infrastructure and Industrial sectors.

The system shall support organizations with more than 70,000 active employees while maintaining excellent performance, scalability and flexibility.

Unlike traditional Payroll systems, this application shall not depend on hardcoded payroll calculations.

Every business calculation shall be configurable.

1.2 System Philosophy

The system shall be designed as a Business Platform rather than a Payroll Program.

Payroll is only one output of the platform.

The platform manages the complete workforce lifecycle.

Employee

↓

Contract

↓

Assignment

↓

Calendar

↓

Shift

↓

Attendance

↓

Timesheet

↓

Payroll

↓

Accounting

↓

Financial Export

↓

Historical Audit

Everything begins with Employee Assignment and ends with Financial Integration.

1.3 Core Business Principles

The following principles shall never change throughout the lifetime of the application.

Principle 1

Business Rules must never be hardcoded.

Principle 2

The system shall support unlimited companies.

Principle 3

The system shall support unlimited projects.

Principle 4

The system shall support unlimited payroll structures.

Principle 5

The system shall support unlimited shifts.

Principle 6

The system shall support unlimited cost codes.

Principle 7

The system shall support unlimited payroll components.

Principle 8

The system shall support unlimited Time Types.

Principle 9

Every calculation must be reproducible years later.

Principle 10

Historical payroll data must never change.

If Labor Law changes today, payroll calculated three years ago must still produce exactly the same result using the historical rule package.

1.4 Scope

The application consists of several integrated business engines.

Human Resources Engine

Contract Engine

Project Management Engine

Calendar Engine

Shift Engine

Attendance Engine

Timesheet Engine

Payroll Engine

Provision Engine

Cost Allocation Engine

Approval Workflow Engine

Notification Engine

Reporting Engine

Export Engine

Rule Engine

Each engine is completely independent.

Communication between engines shall occur only through well-defined business services.

1.5 Target Users

The system shall support the following users.

Human Resources

Payroll Officers

Timekeepers

Data Operators

Project Engineers

Construction Managers

Site Supervisors

Payroll Managers

Finance Department

Management

Employees (Self Service)

External ERP Systems

1.6 Supported Companies

The application shall initially support companies operating under Qatar Labor Law.

Future country support shall include

United Arab Emirates

Saudi Arabia

Oman

Kuwait

Bahrain

Egypt

Jordan

Other countries shall be added by creating new Labor Law Packages.

No software modification shall be required.

1.7 Performance Targets

The application is designed for enterprise organizations.

Minimum target

70,000 Employees

300 Projects

20,000 Cost Codes

5,000 Concurrent Users

3 Million Timesheet Records per Month

Multiple Payroll Runs

Real-time Reporting

Payroll generation shall not exceed ten minutes for 70,000 employees.

1.8 Legacy System

The current Payroll and Timesheet applications contain years of valuable business knowledge.

However,

The legacy database shall never be copied.

The legacy table names shall never be reused.

The legacy stored procedures shall never be migrated.

Only business knowledge shall be transferred.

The new application shall be designed from scratch.

1.9 Technology

Backend

Java 21

Spring Boot

Spring Security

Spring Batch

Spring Scheduler

Spring Data JPA

PostgreSQL

Redis

RabbitMQ

Docker

Kubernetes Ready

Frontend

React

TypeScript

Material UI

AG Grid Enterprise

Authentication

JWT

OAuth2 Ready

REST APIs

Future GraphQL Support

1.10 System Philosophy

The application shall always answer one simple business question.

“How much should this employee receive for this payroll period according to the applicable Labor Law and Company Policy?”

Everything else in the application exists only to answer that question correctly.

1.11 High Level Workflow

Create Company

↓

Create Calendar

↓

Generate Payroll Periods

↓

Create Projects

↓

Create Cost Codes

↓

Create Shifts

↓

Create Rule Package

↓

Create Payroll Components

↓

Create Employees

↓

Create Contracts

↓

Assign Employees to Projects

↓

Assign Employees to Shifts

↓

Generate Monthly Timesheets

↓

Fill Timesheets

↓

Approve Timesheets

↓

Run Payroll

↓

Generate Provisions

↓

Generate Accounting Entries

↓

Export to ERP

↓

Archive Payroll

1.12 System Layers

Presentation Layer

↓

API Layer

↓

Business Layer

↓

Rule Engine

↓

Domain Layer

↓

Persistence Layer

↓

PostgreSQL

No business calculations shall exist inside SQL procedures.

Database responsibility

Store Data

Maintain Integrity

Provide Performance

Application responsibility

Business Rules

Payroll Calculations

Timesheet Calculations

Leave Calculations

Provision Calculations

1.13 Design Goals

The system shall be

Fast

Simple

Maintainable

Scalable

Secure

Auditable

Cloud Ready

Highly Configurable

Country Independent

Labor Law Independent

Company Policy Independent

Future Proof

This philosophy shall guide every design decision throughout the project.

End of Chapter 1

Chapter 2

Business Domain Model

2.1 Introduction

Before discussing Payroll, Timesheets or Attendance, it is essential to understand the Business Domain.

The application is not built around Payroll.

It is built around Employees working on Projects according to Contracts under specific Labor Laws and Company Policies.

Payroll is only the financial result of all business transactions.

2.2 Business Domains

The system consists of independent Business Domains.

Each domain owns its own data and business rules.

The primary domains are:

• Company Management

• Organization Structure

• Employee Management

• Contract Management

• Project Management

• Cost Management

• Calendar Management

• Shift Management

• Attendance Management

• Timesheet Management

- Payroll Management
- Provision Management
- Leave Management
- Overtime Management
- Banking Management
- Workflow Management
- Reporting
- Integration

2.3 Company Domain

The Company is the highest level entity inside the system.

Every other entity belongs to a Company.

A Company owns:

Projects

Employees

Calendars

Payroll Policies

Rule Packages

Departments

Cost Codes

Banks

Approval Workflows

One installation may contain multiple companies.

Companies are completely isolated from each other.

2.4 Organization Structure

Each Company contains an Organization Structure.

Company

↓

Business Unit

↓

Division

↓

Department

↓

Section

↓

Team

↓

Employee

The organization hierarchy is used only for reporting and approvals.

Payroll calculations are independent from the organization hierarchy.

2.5 Employee Domain

The Employee is the core entity of the entire system.

Everything revolves around the Employee.

Employee data shall include:

Employee Number

Government ID

Passport

Nationality

Date of Birth

Gender

Religion

Marital Status

Employment Status

Joining Date

Termination Date

Contact Information

Emergency Contact

Bank Information

Visa Information

Medical Insurance

Accommodation

Transportation

Employee Photo

Documents

Employees are never physically deleted.

Historical information must always remain available.

2.6 Contract Domain

Every employee must have at least one Contract.

A Contract defines:

Employment Type

Salary Structure

Basic Salary

Allowances

Benefits

Working Hours

Working Days

Contract Start Date

Contract End Date

Labor Law Package

Company Policy Package

Payroll Frequency

Shift Assignment

Notice Period

EOS Eligibility

Each employee may have multiple contracts.

Only one contract may be active at any point in time.

Contract history shall always be preserved.

2.7 Project Domain

Projects represent the operational units where employees perform work.

Each Project contains:

Project Code

Project Name

Client

Location

Contract Number

Start Date

End Date

Project Manager

Calendar

Cost Codes

Approval Workflow

Project Status

Projects may contain thousands of employees.

Employees may move between projects.

Project history shall remain available forever.

2.8 Employee Assignment

Employees are not permanently attached to projects.

Instead, assignments are maintained.

Each Assignment contains:

Employee

Project

Effective Date

End Date

Department

Section

Supervisor

Default Shift

Default Cost Code

One employee may have multiple assignments throughout his career.

Historical assignments are mandatory.

2.9 Calendar Domain

Calendar is generated yearly.

Example

Calendar 2026

contains

January

February

March

...

December

Every month becomes one Payroll Period.

Example

January 2026

↓

202601

The Calendar defines:

Date

Day Name

Public Holiday

Special Holiday

Shutdown

Payroll Period

Calendar does NOT define Weekly Off.

Weekly Off belongs to Shift.

2.10 Payroll Period

Payroll processing always occurs inside a Period.

Each Period contains:

Start Date

End Date

Payroll Status

Approval Status

Locked Flag

Processing Date

Closing Date

Periods may be reopened only by authorized users.

2.11 Shift Domain

Shift is one of the most important entities.

Employees work according to Shifts.

Each Shift defines:

Shift Name

Shift Type

Start Time

End Time

Break Duration

Working Hours

Allowed Overtime

Maximum Overtime

Grace Period

Night Shift

Weekly Off Days

Holiday Behavior

Friday Behavior

Attendance Rules

Late Rules

Early Exit Rules

Each employee may change shifts many times during employment.

Shift history shall always remain available.

2.12 Attendance Domain

Attendance records describe when employees actually worked.

Attendance sources include:

Biometric Devices

Excel Import

Crew Attendance

Manual Entry

API Integration

Attendance itself does not calculate payroll.

Attendance becomes input for Timesheet.

2.13 Timesheet Domain

Timesheet is the official payroll source.

Each month the system generates Timesheets automatically.

Each employee receives one Timesheet for the selected Payroll Period.

The Timesheet initially contains default working hours according to Shift Rules.

Example

Employee

Ahmed

Shift

8 Hours

Month

July

The system automatically generates

31 daily records.

The operator does NOT enter all employees manually.

2.14 Timesheet Generation

When a new Payroll Period is opened:

The system performs the following:

Load Active Employees

↓

Load Active Contracts

↓

Load Active Project Assignments

↓

Load Shift Assignments

↓

Generate Daily Records

↓

Apply Calendar

↓

Apply Holidays

↓

Apply Weekly Off

↓

Generate Monthly Timesheet

No manual work is required.

2.15 Copy Days

Large construction companies normally process thousands of employees.

Entering attendance day by day is impossible.

The system shall provide a Copy Days feature.

Example

Select:

Project

↓

Period

↓

Employees

↓

Working Hours = 8

↓

Apply to Entire Month

The system automatically fills all working days.

The operator then modifies only exceptions.

Examples:

Annual Leave

Sick Leave

Absent

Training

Business Trip

Unpaid Leave

Holiday Work

Weekend Work

This dramatically reduces data entry effort.

2.16 Cost Allocation

Employees may work on multiple Cost Codes during the same day.

Example

Employee

Ahmed

Worked

Concrete

4 Hours

Steel

2 Hours

Electrical

2 Hours

Total

8 Hours

Validation

4 + 2 + 2 = 8

If allocation differs from approved hours,

Payroll processing shall stop.

There shall be no limit on the number of Cost Codes.

The system must never contain fields such as:

CC1

CC2

CC3

CC4

Cost allocations shall be stored as independent transaction records.

2.17 Time Types

Every attendance record belongs to one Time Type.

Examples

Present

Absent

Annual Leave

Sick Leave

Unpaid Leave

Training

Business Trip

Friday

Holiday

Declared Overtime

Undeclared Overtime

Emergency Leave

Compassion Leave

Each Time Type controls:

Payroll

Leave Balance

Provision

Approval

Reporting

Cost Allocation

Future Time Types may be added without software modification.

End of Chapter 2 (Part 1)

Chapter 3

Timesheet Engine

3.1 Introduction

The Timesheet Engine is the operational heart of the Workforce Management System.

The purpose of the Timesheet Engine is to convert workforce attendance into approved payroll transactions while accurately allocating labor cost to Projects and Cost Codes.

The Timesheet Engine does not calculate salaries directly.

Instead, it produces approved working transactions that become the primary input for the Payroll Engine.

Attendance devices, Excel sheets and Crew Attendance sheets are considered data sources only.

The approved Timesheet is the official payroll source.

3.2 Business Philosophy

Large construction companies may employ more than 70,000 employees distributed across hundreds of projects.

Entering attendance manually for every employee on a daily basis is impossible.

Therefore the system follows the following philosophy:

Generate

↓

Copy Default Working Days

↓

Modify Exceptions

↓

Approve

↓

Payroll

The operator should never enter attendance for all employees one by one.

Instead, the operator edits only exceptional cases.

3.3 Timesheet Lifecycle

Every Timesheet passes through the following stages.

Draft

↓

Generated

↓

Copy Days

↓

Daily Adjustments

↓

Cost Allocation

↓

Validation

↓

Supervisor Approval

↓

Payroll Approval

↓

Locked

↓

Archived

Once Payroll is generated, the Timesheet becomes read-only unless reopened by an authorized Payroll Manager.

3.4 Timesheet Generation

When a Payroll Period is opened, the system automatically creates Timesheets.

Generation process:

Load Active Employees

↓

Load Active Contracts

↓

Load Active Assignments

↓

Load Active Shifts

↓

Load Calendar

↓

Generate Daily Records

↓

Apply Holidays

↓

Apply Weekly Off

↓

Ready for Editing

Generation shall support more than 70,000 employees in less than one minute.

3.5 Daily Record Structure

Every generated day represents one independent business transaction.

Each record contains:

Employee

Payroll Period

Date

Project

Shift

Working Hours

Approved Hours

Time Type

Status

Remarks

Approval Status

Cost Allocation

Overtime

Undeclared Overtime

Attachments

Audit Information

Each daily record may be edited independently.

3.6 Copy Days

Copy Days is one of the most important business features.

Instead of entering attendance manually for every employee, the operator selects:

Project

Payroll Period

Employee Group

Shift

Working Hours

Time Type

Then executes Copy Days.

Example:

Project:

Doha Metro

Period:

2026-07

Employees:

520

Working Hours:

8

Time Type:

Present

The system automatically fills every working day for every selected employee.

The operator then edits only exceptions.

This feature reduces manual work by more than 95%.

3.7 Exceptions

Only exceptional records require manual editing.

Examples include:

Annual Leave

Sick Leave

Unpaid Leave

Absent

Business Trip

Training

Holiday Work

Weekend Work

Late Joiner

Employee Transfer

Contract Expiry

Termination

Shift Change

Project Change

Every exception shall automatically trigger recalculation of payroll-related values.

3.8 Crew Attendance

Construction projects usually receive Crew Attendance Sheets from Site Engineers.

Crew Attendance represents attendance for a group rather than individual employees.

Example:

Crew:

Concrete Team A

Employees:

48

Worked Hours:

10

Project:

P001

Cost Code:

CC101

The operator imports the Crew Attendance.

The system distributes attendance automatically to all employees belonging to the Crew.

Employees requiring different attendance are edited individually.

3.9 Excel Import

The Timesheet Engine shall support Excel Import.

Supported fields include:

Employee Number

Date

Hours Worked

Time Type

Declared OT

Undeclared OT

Project

Cost Code

Remarks

The import process shall perform complete validation before saving data.

No invalid data shall be imported.

3.10 Work Segments

A single employee may work on multiple Projects or Cost Codes during the same day.

Instead of storing one attendance record per day, the system shall support multiple Work Segments.

Example

Employee:

Ahmed

Date:

15-Jul-2026

Segment 1

Project:

Project A

Cost Code:

Concrete

Hours:

4

Segment 2

Project:

Project B

Cost Code:

Steel

Hours:

2

Segment 3

Project:

Project C

Cost Code:

Finishing

Hours:

2

Total Approved Hours = 8

This design eliminates limitations such as CC1, CC2, CC3 used in legacy systems.

The number of Work Segments shall be unlimited.

3.11 Cost Allocation Validation

Cost Allocation shall always be validated before approval.

Validation Rules:

Total Cost Allocation Hours

=

Approved Working Hours

Example

Concrete

4

Steel

2

Electrical

2

Total

8

Approved Hours

8

Validation Passed

Example

Concrete

5

Steel

3

Total

8

Approved Hours

10

Validation Failed

Payroll processing shall stop until validation succeeds.

3.12 Shift Changes

Employees may change shifts during the same Payroll Period.

Example

01-Jul to 10-Jul

Morning Shift

11-Jul to 20-Jul

Night Shift

21-Jul to 31-Jul

Ramadan Shift

Each day shall always use the Shift assigned on that specific date.

Historical Shift information shall never be modified.

3.13 Project Transfers

Employees may transfer between projects.

Example

01-Jul to 12-Jul

Project A

13-Jul to 31-Jul

Project B

Payroll shall calculate labor cost according to the Project active on each transaction date.

Historical Project Assignments shall remain unchanged.

3.14 Overtime inside Timesheet

The Timesheet stores working hours.

It does not determine payment.

Example

Worked Hours

12

Approved Shift Hours

8

Declared Overtime

2

Undeclared Overtime

2

The Payroll Engine later determines how each hour will be paid according to Labor Law and Company Policy.

3.15 Undeclared Overtime

Undeclared Overtime represents worked hours exceeding approved overtime.

Example

Shift

8 + 2 Approved OT

Worked

13 Hours

Result

Normal

8

Declared OT

2

Undeclared OT

3

Undeclared OT shall be stored independently.

It may be:

Rejected

Approved

Paid Outside Payroll

Converted into Future Payment

Converted into Time Off

The calculation depends on Company Policy.

3.16 Approval Workflow

Each Timesheet passes through configurable approval stages.

Typical workflow:

Timekeeper

↓

Supervisor

↓

Project Manager

↓

Payroll Department

↓

Finance (Optional)

Approval levels shall be configurable.

Different companies may define different workflows.

3.17 Audit Trail

Every Timesheet modification shall be recorded.

Audit Information includes:

Previous Value

New Value

User

Date

Reason

Approval History

The system shall never permanently delete Timesheet transactions.

Historical audit data shall remain available indefinitely.

3.18 Performance Requirements

The Timesheet Engine shall support:

70,000+ Employees

Millions of Daily Records

Concurrent Editing

Bulk Copy Days

Bulk Excel Import

Crew Attendance Import

Bulk Cost Allocation

Bulk Approval

The target generation time for monthly Timesheets shall not exceed one minute under normal operating conditions.

3.19 Business Rules

The Timesheet Engine shall not contain payroll calculations.

Its responsibility is limited to:

Recording Work

Validating Work

Approving Work

Allocating Work

Producing Approved Working Transactions

All salary calculations shall be performed exclusively by the Payroll Engine.

End of Chapter 3

Chapter 4

Calendar & Shift Engine

4.1 Introduction

The Calendar & Shift Engine is responsible for defining the official working schedule of the organization.

It is the foundation of the entire Workforce Management System.

Every Timesheet, Attendance transaction and Payroll calculation depends on the Calendar and Shift configuration.

The Calendar determines **when work is expected**.

The Shift determines **how work should be performed**.

Neither Payroll nor Timesheet calculations may bypass these two modules.

4.2 Business Philosophy

The Calendar does NOT belong to Employees.

The Calendar belongs to the Company.

The Shift belongs to Employees through Assignments.

This separation allows:

- Multiple Calendars
- Multiple Shifts
- Multiple Companies
- Different Holidays
- Different Working Schedules
- Future Labor Law Changes

without changing Payroll logic.

4.3 Calendar Hierarchy

The Calendar hierarchy is defined as follows:

Company

↓

Calendar

↓

Calendar Year

↓

Payroll Period

↓

Calendar Day

Each level is generated automatically.

4.4 Calendar Creation

A Calendar is normally created once per year.

Example:

Calendar

2026

The system automatically generates:

January

February

March

April

May

June

July

August

September

October

November

December

Each month becomes one Payroll Period.

4.5 Payroll Period

Every Payroll Period contains:

Period Code

Example

202601

Period Name

January 2026

Start Date

End Date

Working Days

Calendar Days

Payroll Status

Approval Status

Closing Date

Lock Status

Periods are independent.

Closing one Period shall not affect another Period.

4.6 Calendar Day

Every day inside the Calendar stores business information.

Each Calendar Day contains:

Date

Day Name

Month

Year

Payroll Period

Public Holiday Flag

Special Holiday Flag

Shutdown Flag

Description

Remarks

Calendar Days do NOT define Weekly Off.

Weekly Off is determined by Shift.

4.7 Public Holidays

Public Holidays are configured once inside the Calendar.

Example:

National Day

Eid Al Fitr

Eid Al Adha

National Sports Day

Founding Day

Company Holiday

Every Holiday contains:

Holiday Name

Holiday Date

Holiday Type

Paid Flag

Country

Description

Applicable Companies

Applicable Projects (Optional)

Holiday calculations shall be delegated to the Rule Engine.

4.8 Shift Philosophy

Employees never work directly according to the Calendar.

Employees work according to Shifts.

The Calendar only defines dates.

The Shift defines working behavior.

4.9 Shift Structure

Each Shift contains:

Shift Code

Shift Name

Shift Category

Start Time

End Time

Break Start

Break End

Break Duration

Working Hours

Allowed Declared OT

Maximum OT

Night Shift Flag

Flexible Shift Flag

Grace Period

Late Tolerance

Early Exit Tolerance

Attendance Calculation Method

Status

Effective Date

Expiry Date

4.10 Weekly Off

Weekly Off belongs to Shift.

Example:

Shift A

Friday

Shift B

Saturday

Shift C

Friday + Saturday

Shift D

Rotational

This allows employees working in different industries to follow different work schedules.

4.11 Shift Assignment

Employees are assigned to Shifts.

Assignments are date driven.

Example

Ahmed

01-Jan

↓

Morning Shift

15-May

↓

Night Shift

01-Aug

↓

Ramadan Shift

Historical assignments shall never be deleted.

Payroll shall always use the Shift effective on the transaction date.

4.12 Shift Versioning

Shifts may evolve over time.

Instead of modifying an existing Shift,

a new Shift Version shall be created.

Example:

Morning Shift V1

Working Hours

8

Allowed OT

2

Effective

01-Jan-2026

Morning Shift V2

Working Hours

9

Allowed OT

1

Effective

01-Jan-2027

Historical Payroll calculations shall continue using Version 1.

4.13 Shift Types

The system shall support unlimited Shift Types.

Examples:

Morning Shift

Evening Shift

Night Shift

Ramadan Shift

Split Shift

Flexible Shift

Rotational Shift

Camp Shift

Remote Shift

Future Shift Types may be added without software modification.

4.14 Attendance Window

Each Shift defines an attendance window.

Example

Shift Start

06:00

Grace Period

15 Minutes

Attendance Accepted Until

06:15

Late Arrival begins after

06:15

Early Arrival may also be configured.

Attendance calculation shall use these limits.

4.15 Break Rules

Every Shift defines Break Rules.

Example

Lunch Break

12:00

↓

13:00

Paid Break

Yes

Or

No

Multiple Breaks shall be supported.

Example:

Breakfast

Tea

Lunch

Prayer

Custom Break

Break calculations shall remain configurable.

4.16 Working Hours

Each Shift defines:

Standard Hours

Maximum Hours

Minimum Hours

Declared OT Limit

Undeclared OT Limit

Night Hours

Holiday Hours

Friday Hours

These values become input to the Timesheet Engine.

4.17 Overtime Rules

Each Shift defines overtime limitations.

Example

Normal Hours

8

Declared OT

2

Maximum Working Hours

12

Employee Worked

13

Result

Normal

8

Declared OT

2

Undeclared OT

3

The Payroll Engine later determines payment.

4.18 Ramadan Support

The system shall support special seasonal shifts.

Example

Ramadan Shift

Working Hours

6

Weekly Off

Friday

Declared OT

2

Effective Dates

01-Mar

↓

30-Mar

Assignments shall automatically return to normal after Ramadan ends.

4.19 Rotational Shifts

Some industries operate using rotational schedules.

Example

4 Days Work

↓

2 Days Off

Or

14 Days Work

↓

7 Days Off

The Shift Engine shall support custom rotation patterns.

4.20 Holiday Behavior

Working on Public Holidays depends on Company Policy and Labor Law.

Example

Holiday

↓

Employee did NOT work

↓

Paid normally

Holiday

↓

Employee worked

↓

Holiday OT

↓

Calculated later by Payroll Engine

The Shift Engine only classifies the day.

The Payroll Engine performs the calculation.

4.21 Friday Behavior

Friday calculations vary by country and company.

Example

Employee did not work

↓

Normal Friday

Employee worked

↓

Friday Overtime

The Rule Engine determines payment.

4.22 Shift Validation

Before assigning a Shift:

The system validates:

Shift Exists

Shift Active

Effective Date Valid

Assignment Does Not Overlap

Calendar Exists

Company Matches

Project Matches (Optional)

Invalid assignments shall be rejected.

4.23 Performance Requirements

The Shift Engine shall support:

Unlimited Shifts

Unlimited Shift Versions

Unlimited Assignments

Historical Queries

Bulk Assignment

Bulk Shift Change

Automatic Assignment

Shift generation and assignment operations shall complete within seconds even for tens of thousands of employees.

4.24 Business Responsibilities

The Calendar Engine is responsible for:

Managing Years

Managing Payroll Periods

Managing Holidays

Managing Calendar Days

The Shift Engine is responsible for:

Working Hours

Weekly Off

Breaks

Attendance Windows

Overtime Limits

Shift Versions

Employee Shift Assignments

Neither module performs payroll calculations.

They provide validated business data consumed by the Timesheet Engine and Payroll Engine.

End of Chapter 4

Chapter 5

Time Type Engine

5.1 Introduction

The Time Type Engine is one of the most critical modules within the Workforce Management & Payroll System.

Unlike traditional payroll systems where attendance types are represented as simple codes, this system treats every Time Type as a complete Business Rule.

A Time Type does not simply identify attendance.

It defines how that attendance affects Payroll, Leave, Overtime, Provisions, Reporting, Cost Allocation and Employee Statistics.

The Time Type Engine is fully configurable.

No Time Type shall require software modification.

5.2 Philosophy

Time Types represent business events.

Examples include:

Present

Annual Leave

Sick Leave

Unpaid Leave

Absent

Training

Business Trip

Public Holiday

Weekend

Declared Overtime

Undeclared Overtime

Emergency Leave

Marriage Leave

Compassion Leave

Work From Home

Suspension

Termination

Future Time Types

Every Time Type is an independent business object.

5.3 Time Type Categories

The system groups Time Types into categories.

Attendance

Leave

Holiday

Overtime

Business

Training

Administrative

Penalty

Future Categories

Categories exist for reporting only.

Business Rules remain independent.

5.4 Time Type Structure

Each Time Type shall contain:

Code

Name

Description

Category

Color

Icon

Active

Effective Date

Expiry Date

Labor Law Package

Company Policy Package

Approval Workflow

Priority

Status

Remarks

5.5 Payroll Behavior

Every Time Type determines how Payroll behaves.

Examples

Present

↓

Normal Salary

Annual Leave

↓

Paid Salary

Unpaid Leave

↓

Salary Deduction

Sick Leave

↓

Depends on Labor Law

Business Trip

↓

Normal Salary

Holiday Work

↓

Holiday Overtime

Declared Overtime

↓

OT Payment

Undeclared Overtime

↓

Pending Decision

Time Types never calculate money.

They determine Payroll Behavior.

5.6 Payroll Components

Each Time Type controls every Payroll Component individually.

Example

Annual Leave

Basic Salary

Paid

Housing

Paid

Transportation

Paid

Food

Paid

Car Allowance

Paid

Schooling

Not Paid

Performance Bonus

Not Paid

Attendance Bonus

Not Paid

EOS

Included

Air Ticket

Included

Another example

Unpaid Leave

Basic Salary

Deduct

Housing

Deduct

Transportation

Deduct

Food

Deduct

Schooling

No Payment

Bonus

No Payment

EOS

Depends on Labor Law

Provision

Deduct

The system shall support unlimited Payroll Components.

5.7 Leave Balance Behavior

Every Leave Time Type defines how Leave Balance behaves.

Examples

Annual Leave

↓

Consume Balance

Sick Leave

↓

Separate Balance

Business Trip

↓

No Leave Balance

Training

↓

No Leave Balance

Compassion Leave

↓

Separate Balance

Future Leave Types

↓

Configurable

5.8 Overtime Behavior

Every Time Type defines overtime behavior.

Examples

Holiday Work

↓

Holiday OT

Friday Work

↓

Friday OT

Night Work

↓

Night OT

Declared OT

↓

Payroll

Undeclared OT

↓

Pending

Training

↓

No OT

Business Trip

↓

Company Policy

5.9 Approval Workflow

Some Time Types require approval.

Examples

Annual Leave

Supervisor

↓

HR

↓

Payroll

Business Trip

Supervisor

↓

Project Manager

↓

HR

Undeclared OT

Supervisor

↓

Construction Manager

↓

Payroll

Approval Workflow shall be configurable.

5.10 Provision Behavior

Every Time Type defines Provision behavior.

Example

Annual Leave

EOS

Included

Leave Provision

Included

Air Ticket

Included

Unpaid Leave

EOS

Deduct

Leave Provision

Deduct

Air Ticket

Depends on Policy

5.11 Reporting Behavior

Every Time Type determines where it appears.

Examples

Payroll Report

Attendance Report

Project Report

Cost Report

Leave Report

Executive Dashboard

Employee Self Service

The same Time Type may appear in multiple reports.

5.12 Cost Allocation

Not every Time Type requires Cost Allocation.

Examples

Present

↓

Mandatory

Business Trip

↓

Optional

Annual Leave

↓

Not Required

Sick Leave

↓

Not Required

Training

↓

Optional

Holiday Work

↓

Mandatory

The behavior shall be configurable.

5.13 Validation Rules

Each Time Type may contain validation.

Examples

Annual Leave

Employee Balance Required

Unpaid Leave

No Balance Required

Business Trip

Travel Request Required

Training

Training Approval Required

Holiday Work

Holiday Must Exist

Declared OT

OT Authorization Required

Validation Rules shall be configurable.

5.14 Time Type Versioning

Business Rules change.

Instead of modifying a Time Type,

the system creates a new version.

Example

Annual Leave V1

↓

Annual Leave V2

↓

Annual Leave V3

Historical Payroll always references the version effective at the transaction date.

5.15 Time Type Assignment

Time Types may be restricted.

Examples

Construction Employees

Office Employees

Daily Workers

Monthly Employees

Temporary Employees

Managers

Company Specific

Project Specific

Country Specific

Restrictions are configurable.

5.16 Business Rule Integration

Each Time Type communicates with:

Calendar Engine

Shift Engine

Payroll Engine

Leave Engine

Provision Engine

Reporting Engine

Rule Engine

Notification Engine

Workflow Engine

No module calculates Time Types independently.

5.17 Future Expansion

Future Time Types shall require only configuration.

No software modification.

No database modification.

No API modification.

Only configuration.

5.18 Responsibilities

The Time Type Engine is responsible for:

Attendance Classification

Payroll Behavior

Leave Behavior

Provision Behavior

Approval Workflow

Reporting Behavior

Validation

Rule Versioning

The Time Type Engine is NOT responsible for salary calculation.

Payroll calculations belong exclusively to the Payroll Engine.

End of Chapter 5

Chapter 6

Payroll Component Engine

6.1 Introduction

The Payroll Component Engine defines every financial element that may affect an employee’s salary.

Unlike traditional payroll systems where salary is stored in fixed columns, this system treats every salary element as an independent Payroll Component.

Examples:

Basic Salary

Housing Allowance

Transportation Allowance

Food Allowance

Mobile Allowance

Car Allowance

Site Allowance

Risk Allowance

Performance Bonus

Attendance Bonus

Schooling Allowance

Air Ticket

Commission

Overtime

Loans

Penalties

Deductions

EOS Provision

Leave Provision

Future Components

The system shall support unlimited Payroll Components.

6.2 Philosophy

Salary is NOT a fixed structure.

Salary is simply a collection of Payroll Components.

Example

Ahmed

Basic Salary

4,500

Housing

2,000

Transportation

500

Food

300

Site Allowance

200

Performance Bonus

800

Loan Deduction

-400

Net Salary

Calculated

No component is special.

Every component follows exactly the same engine.

6.3 Component Categories

Each Payroll Component belongs to one category.

Examples

Salary

Allowance

Deduction

Provision

Bonus

Overtime

Leave

Loan

Government

Tax

Insurance

Future Categories

Categories are used for reporting and grouping only.

6.4 Component Definition

Every Payroll Component contains:

Component Code

Component Name

Category

Description

Status

Effective Date

Expiry Date

Currency

Calculation Method

Rounding Method

Priority

Visible on Payslip

Visible on Reports

Taxable

WPS Included

EOS Included

Provision Included

Leave Included

Cost Allocation Required

Approval Required

Remarks

6.5 Payment Frequency

Every Payroll Component defines when it is paid.

Supported frequencies include:

Every Payroll

Weekly

Biweekly

Monthly

Quarterly

Semi Annual

Annual

One Time

Manual

Example

Basic Salary

Monthly

Schooling

Every 6 Months

Bonus

Yearly

Air Ticket

Every 24 Months

Commission

Manual

6.6 Calculation Method

Every component defines how its value is calculated.

Supported methods include:

Fixed Amount

Percentage

Formula

Hourly Rate

Daily Rate

Monthly Rate

Contract Value

Rule Engine

Manual Entry

Future Calculation Methods

Example

Housing

40%

×

Basic Salary

Performance Bonus

Manual

Attendance Bonus

Rule Engine

6.7 Calculation Priority

Payroll Components are processed sequentially.

Example

Basic Salary

↓

Housing

↓

Transportation

↓

Overtime

↓

Bonus

↓

Loans

↓

Penalties

↓

Tax

↓

Net Salary

Priority determines dependency.

6.8 Payroll Component Matrix

Every Time Type controls every Payroll Component.

Example

Annual Leave

Basic Salary

Paid

Housing

Paid

Transportation

Paid

Food

Paid

Mobile

Paid

Car

Paid

Schooling

Not Paid

Performance Bonus

Not Paid

Attendance Bonus

Not Paid

EOS

Included

Leave Provision

Included

Air Ticket

Included

Example

Unpaid Leave

Basic Salary

Deduct

Housing

Deduct

Transportation

Deduct

Food

Deduct

Car

Deduct

Attendance Bonus

Deduct

EOS

Depends on Labor Law

Leave Provision

Deduct

Air Ticket

Depends on Policy

This matrix is configurable.

No software modification is required.

6.9 Formula Components

Some Payroll Components depend on other components.

Example

Housing

40%

×

Basic Salary

Food

10%

×

Basic Salary

EOS

Basic Salary

Housing

Transportation

The Formula Engine resolves dependencies automatically.

6.10 Manual Components

Some components are entered manually.

Examples

Performance Bonus

Adjustment

Salary Correction

Manual Penalty

Manual Reward

Manual entries remain fully auditable.

6.11 Recurring Components

Recurring Components are generated automatically.

Examples

Basic Salary

Housing

Transportation

Car Allowance

Mobile

Recurring Components continue until expiry.

6.12 Scheduled Components

Some Payroll Components are paid according to schedules.

Example

Schooling

Every 6 Months

Air Ticket

Every 24 Months

Annual Bonus

Every December

The Payroll Engine automatically activates the component during the correct Payroll Period.

6.13 One-Time Components

Examples

Salary Adjustment

Joining Bonus

Retention Bonus

Special Reward

Settlement Adjustment

Paid once only.

6.14 Deductions

Deductions are Payroll Components.

Examples

Loan

Penalty

Salary Advance

Court Order

Insurance

Accommodation Recovery

Telephone Charges

Traffic Violations

Medical Recovery

Future deductions

No special logic exists for deductions.

6.15 Provisions

Provision Components are not necessarily paid to employees.

Examples

EOS Provision

Leave Provision

Air Ticket Provision

Provision Components affect Accounting but may not appear on Payslips.

6.16 Cost Allocation

Each Payroll Component defines cost allocation behavior.

Examples

Basic Salary

Allocated

Housing

Allocated

Bonus

Allocated

Loan

Not Allocated

EOS Provision

Allocated

Tax

Not Allocated

6.17 Approval

Certain Payroll Components require approval.

Examples

Bonus

Loan

Salary Adjustment

Manual Overtime

Retention Bonus

Approval workflows are configurable.

6.18 Payslip Behavior

Every Payroll Component defines:

Show on Payslip

Hide from Payslip

Merge with another Component

Internal Only

Examples

EOS Provision

Hidden

Basic Salary

Visible

Loan

Visible

Accounting Adjustment

Hidden

6.19 Historical Versioning

Payroll Components shall never be modified historically.

Instead,

Version 1

↓

Version 2

↓

Version 3

Payroll always references the version active during payroll calculation.

6.20 Business Responsibilities

The Payroll Component Engine is responsible for:

Component Definition

Calculation Rules

Formula Resolution

Payment Frequency

Versioning

Reporting

Approval

Cost Allocation

Integration

The Payroll Component Engine never calculates the final Payroll.

That responsibility belongs exclusively to the Payroll Engine.

End of Chapter 6

Chapter 7 - Payroll Engine

7.1 Introduction
The Payroll Engine is the financial core of the Workforce Management & Payroll System. Its responsibility is to transform approved Timesheets into payroll transactions while applying Labor Law, Company Policy, Contracts, Payroll Components and Time Type rules.

7.2 Payroll Objectives

- Process Monthly Employees
- Process Daily Workers
- Support Weekly, Monthly, Off-Cycle and Final Settlement Payroll
- Support Retroactive Payroll
- Generate Payslips, Journal Entries, Bank/WPS Files

7.3 Payroll Inputs

Employee
Contract
Assignment
Approved Timesheet
Calendar
Shift
Payroll Components
Time Types
Additions & Deductions
Leave
Provision Rules

7.4 Payroll Processing Flow

Open Payroll Period
↓
Load Employees
↓
Validate Employee
↓
Load Contract
↓
Load Approved Timesheet
↓
Apply Time Type Rules
↓
Apply Shift Rules
↓
Apply Labor Law
↓
Apply Company Policy
↓
Calculate Earnings
↓
Calculate Deductions
↓
Calculate Provisions
↓
Generate Gross Salary
↓
Generate Net Salary
↓
Generate Payslip
↓
Generate Journal Entries
↓
Generate Bank/WPS Files

7.5 Payroll Validation

Validate:
• Employee Active
• Active Contract
• Approved Timesheet
• Valid Assignment
• Open Payroll Period
• Balanced Cost Allocation

• Rule Package Exists
Invalid employees are skipped and logged.

7.6 Business Scenario 1 - Monthly Employee

Employee: Ahmed Hassan
Basic Salary: 5,000 QAR
Housing: 2,000
Transport: 500
Food: 300
Attendance Bonus: 200

Timesheet:
26 Present Days
2 Annual Leave
1 Holiday Worked
4 Declared OT Hours

Gross Salary:

Basic	5,000
Housing	2,000
Transport	500
Food	300
Attendance Bonus	200
Holiday OT	350

Gross	8,350
Loan	-500
Medical	-150

Net Salary	7,700 QAR

7.7 Business Scenario 2 - Daily Worker

Daily Rate: 150 QAR
Worked Days: 25
Absent: 3

Salary = 25 × 150 = 3,750
Deduction = 3 × 150 = 450
Net Salary = 3,300 QAR

7.8 Business Scenario 3 - Multiple Cost Codes

15 July
Project A / Concrete : 4 Hours
Project A / Steel : 2 Hours
Project B / Electrical:2 Hours

Approved Hours = 8
Cost Allocation = Valid
Payroll = 8 Hours

7.9 Business Scenario 4 - Salary Change

Old Basic Salary : 5,000
Effective Date : 15 July
New Basic Salary : 6,000

Payroll automatically prorates:
1-14 July -> Old Salary
15-31 July -> New Salary

7.10 Retroactive Payroll

If a salary increase is approved later, the engine creates adjustment transactions for previous periods without modifying historical payroll.

7.11 Performance Requirements

Support:
• 70,000+ Employees
• Parallel Processing
• Multiple Companies
• Millions of Transactions
Target Payroll Execution: < 10 Minutes

7.12 Outputs

Employee Payslip
Payroll Register
Cost Distribution
Journal Entries
Bank Transfer File
WPS File
ERP Export
Payroll Audit Log

Chapter 8

Leave Management Engine

8.1 Introduction

The Leave Management Engine is responsible for managing every employee leave transaction from the beginning of employment until Final Settlement.

The Leave Engine is fully integrated with:

- Employee Management
- Contract Management
- Calendar Engine
- Shift Engine
- Payroll Engine
- Time Type Engine
- Provision Engine
- Rule Engine
- Workflow Engine

Every leave transaction affects one or more of these modules.

The Leave Engine does not calculate salary directly.

It produces approved leave transactions consumed by the Payroll Engine.

8.2 Business Philosophy

Leave is not simply an employee absence.

A Leave Transaction is a legal and financial event.

Every leave shall answer the following questions:

- Is approval required?
- Is the leave paid?
- Which Payroll Components are paid?
- Does it reduce Leave Balance?

- Does it generate Leave Provision?
- Does it affect EOS?
- Does it require supporting documents?
- Does Labor Law define special rules?
- Does Company Policy override Labor Law?

All these answers shall be configurable.

8.3 Leave Types

The system shall support unlimited Leave Types.

Examples include:

Annual Leave

Sick Leave

Emergency Leave

Compassion Leave

Marriage Leave

Maternity Leave

Paternity Leave

Study Leave

Training Leave

Business Trip

Work From Home

Unpaid Leave

Special Leave

Suspension

Government Leave

Future Leave Types

No software modification shall be required to create a new Leave Type.

8.4 Leave Categories

Leave Types are grouped into categories.

Examples

Paid Leave

Unpaid Leave

Medical Leave

Special Leave

Business Leave

Government Leave

Administrative Leave

Categories are used for reporting purposes only.

Business Rules remain independent.

8.5 Leave Configuration

Every Leave Type contains:

Leave Code

Leave Name

Category

Description

Active Status

Effective Date

Expiry Date

Approval Workflow

Rule Package

Company Policy

Labor Law Package

Payroll Matrix

Leave Balance Rule

Provision Rule

Required Attachments

Maximum Duration

Minimum Duration

Carry Forward Rule

Negative Balance Allowed

Future Leave Types shall require configuration only.

8.6 Leave Balance

Each employee has independent Leave Balances.

Examples

Annual Leave

30 Days

Sick Leave

15 Days

Emergency Leave

5 Days

Compassion Leave

3 Days

Maternity Leave

90 Days

Every Leave Balance is maintained separately.

8.7 Leave Accrual

Leave Balance may be generated using different methods.

Supported methods include:

Monthly Accrual

Annual Allocation

Daily Accrual

Contract Based

Manual Allocation

Rule Engine

Example

Annual Leave

30 Days

Monthly Accrual

2.5 Days per Month

After four months:

2.5 × 4 = 10 Days

Available Balance

10 Days

8.8 Leave Request

Employees may request Leave through:

HR Department

Employee Self Service

Mobile Application

Supervisor

Imported Request

Every request contains:

Employee

Leave Type

Start Date

End Date

Total Days

Reason

Attachments

Comments

Workflow Status

8.9 Leave Approval Workflow

Approval shall be configurable.

Example

Employee

↓

Supervisor

↓

Project Manager

↓

HR Department

↓

Payroll

Different Leave Types may have different workflows.

8.10 Leave Validation

Before Leave Approval the system validates:

Employee Active

Contract Exists

Leave Balance Available

No Date Overlap

Payroll Period Open

Supporting Documents Available

Approval Authority Exists

Rule Package Exists

If validation fails,

Leave shall not be approved.

8.11 Paid Leave Rules

Paid Leave is controlled by the Payroll Component Matrix.

Example

Annual Leave

Basic Salary → Paid

Housing → Paid

Transport → Paid

Food → Paid

Attendance Bonus → No

Performance Bonus → Depends on Company Policy

EOS → Included

Leave Provision → Included

Air Ticket → Included

8.12 Unpaid Leave Rules

Example

Unpaid Leave

Basic Salary → Deduct

Housing → Deduct

Transport → Deduct

Food → Deduct

Attendance Bonus → Deduct

EOS → Depends on Labor Law

Provision → Deduct

Air Ticket → Depends on Policy

8.13 Leave Calendar

The Leave Engine uses the Calendar Engine.

Example

Annual Leave

Start

01 August

End

10 August

Public Holidays

Automatically detected.

Weekly Off

Automatically detected according to the assigned Shift.

The Leave Duration is calculated according to the configured Rule Package.

8.14 Leave Payroll Integration

Approved Leave transactions are transferred automatically to Payroll.

Payroll determines payment using:

Time Type Rules

Payroll Component Matrix

Labor Law

Company Policy

No manual Payroll adjustment is required.

8.15 Leave Encashment

Employees may convert Leave Balance into cash.

Example

Available Leave

20 Days

Requested Encashment

10 Days

Payroll automatically generates

Leave Encashment Component

Remaining Leave Balance

10 Days

The calculation method is configurable.

8.16 Leave Settlement

When an employee resigns or is terminated, the Leave Engine calculates:

Unused Leave

Paid Leave

Unpaid Leave

Leave Encashment

Payroll receives the result automatically.

8.17 Business Scenario 1

Employee

Ahmed

Annual Leave Balance

24 Days

Requested

10 Days

Approved

Yes

Remaining Balance

14 Days

Payroll

Basic Salary → Paid

Housing → Paid

Transport → Paid

Attendance Bonus → Not Paid

8.18 Business Scenario 2

Employee

Mohammed

Unpaid Leave

5 Days

Basic Salary

5,000

Daily Rate

166.67

Payroll Deduction

5 × 166.67

=

833.35

Housing and Transportation deducted according to Company Policy.

8.19 Business Scenario 3

Employee

Fatima

Sick Leave

7 Days

Medical Certificate

Attached

Approval

Automatic

Payroll

Calculated according to Qatar Labor Law Rule Package.

8.20 Audit Trail

Every Leave transaction records:

Employee

Leave Type

Old Balance

New Balance

Approval History

Supporting Documents

User

Date

Reason

Historical Leave transactions shall never be deleted.

8.21 Performance Requirements

The Leave Engine shall support:

Unlimited Leave Types

Unlimited Leave Rules

Unlimited Approval Workflows

70,000+ Employees

Millions of Leave Transactions

Real-Time Leave Balance

Bulk Leave Processing

8.22 Business Responsibilities

The Leave Engine is responsible for:

Leave Requests

Leave Balance

Leave Accrual

Leave Approval

Leave Validation

Leave Encashment

Leave Settlement

Payroll Integration

Provision Integration

Audit Trail

The Leave Engine does NOT calculate salaries.

Salary calculations belong exclusively to the Payroll Engine.

End of Chapter 8

Chapter 9

Overtime Management Engine

9.1 Introduction

The Overtime Management Engine is responsible for managing all employee overtime transactions.

Unlike traditional payroll systems where overtime is entered directly into Payroll, this system calculates overtime through an independent engine before Payroll processing.

The Overtime Engine is integrated with:

- Calendar Engine
- Shift Engine
- Timesheet Engine
- Payroll Engine

- Rule Engine
- Approval Workflow
- Cost Allocation Engine

9.2 Business Philosophy

Working extra hours does not necessarily mean that those hours will be paid.

Every overtime transaction passes through three stages:

Worked

↓

Validated

↓

Approved

↓

Paid

Therefore,

Worked Hours

≠

Approved Overtime

≠

Paid Overtime

9.3 Overtime Types

The system shall support unlimited overtime types.

Examples

Declared Overtime

Undeclared Overtime

Holiday Overtime

Friday Overtime

Weekend Overtime

Night Overtime

Emergency Overtime

Travel Overtime

Site Overtime

Standby Overtime

Call-Out Overtime

Future Overtime Types

9.4 Declared Overtime

Declared Overtime represents overtime approved before or during execution.

Example

Shift

8 Hours

Approved OT

2 Hours

Worked

10 Hours

Payroll

8 Normal

2 Declared OT

9.5 Undeclared Overtime

Undeclared Overtime represents worked hours exceeding the approved overtime.

Example

Shift

8 + 2

Worked

13

Result

Normal

8

Declared OT

2

Undeclared OT

3

Undeclared OT shall remain pending until management decision.

9.6 Overtime Authorization

Certain overtime requires authorization before payment.

Example Workflow

Employee

↓

Supervisor

↓

Project Manager

↓

Construction Manager

↓

Payroll

The workflow is configurable.

9.7 Shift Rules

Every Shift defines:

Standard Hours

Declared OT Limit

Maximum Daily Hours

Night Hours

Holiday Rules

Friday Rules

The Overtime Engine shall always use the active Shift.

9.8 Holiday Overtime

Working during Public Holidays follows Labor Law.

Example

Holiday

Employee Worked

10 Hours

Shift

8 Hours

Result

8 Holiday Hours

2 Holiday OT

Payroll calculation depends on Labor Law Package.

9.9 Friday Overtime

Working during Weekly Off is treated separately.

Example

Weekly Off

Friday

Worked

8 Hours

Payroll

Friday Overtime

The payment method depends on Company Policy and Labor Law.

9.10 Night Overtime

Night Hours may have special rates.

Example

Night Shift

22:00

↓

06:00

Worked

10 Hours

Night Premium

Configured by Rule Package.

9.11 Overtime Approval

Each overtime transaction may have one of the following statuses:

Draft

Submitted

Approved

Rejected

Paid

Cancelled

Historical records shall never be deleted.

9.12 Overtime Payroll Integration

Approved overtime transactions are transferred automatically to Payroll.

Payroll determines payment according to:

Labor Law

Company Policy

Payroll Component Matrix

Rule Engine

9.13 Business Scenario 1

Employee

Ahmed

Shift

8 + 2

Worked

11 Hours

Payroll Result

Normal

8

Declared OT

2

Undeclared OT

1

9.14 Business Scenario 2

Employee

Mohammed

Worked on National Holiday

10 Hours

Payroll

Holiday OT

Calculated automatically according to Qatar Labor Law.

9.15 Business Scenario 3

Employee

Fatima

Worked Friday

8 Hours

Company Policy

Pay Friday OT at 150%

Payroll automatically generates Friday OT component.

9.16 Business Scenario 4

Employee

Ali

Worked

15 Hours

Shift

8 + 2

Result

Normal

8

Declared OT

2

Undeclared OT

5

Supervisor rejected

2 Hours

Approved

3 Hours

Payroll pays only approved hours.

9.17 Validation

The Overtime Engine validates:

Employee Active

Shift Exists

Approved Timesheet

Working Hours Valid

Daily Limit

Labor Law Package

Approval Authority

Project Assignment

9.18 Audit Trail

Every overtime transaction stores:

Worked Hours

Approved Hours

Rejected Hours

Approval History

Rule Version

User

Date

Reason

9.19 Performance Requirements

Support

70,000 Employees

Millions of Overtime Transactions

Bulk Approval

Bulk Import

Real-Time Calculation

Parallel Processing

9.20 Responsibilities

The Overtime Engine is responsible for:

Capturing Overtime

Validating Overtime

Approving Overtime

Integrating with Payroll

Maintaining Audit History

The Payroll Engine is responsible only for paying approved overtime.

End of Chapter 9

Chapter 10

Provision Management Engine

10.1 Introduction

The Provision Management Engine is responsible for calculating and maintaining all employee accruals throughout the employment lifecycle.

Unlike Payroll, which calculates amounts payable to employees, the Provision Engine calculates future liabilities that the company must recognize for accounting purposes.

Provision calculations are fully integrated with:

- Payroll Engine
- Leave Engine
- Contract Management
- Accounting Module
- Rule Engine
- Labor Law Package
- Company Policy

The Provision Engine never generates employee payments directly.

Its responsibility is to calculate company liabilities.

10.2 Business Philosophy

Every month the company owes employees certain future benefits.

Although these benefits are not paid immediately, they must be recognized financially.

Examples include:

End of Service (EOS)

Leave Provision

Air Ticket Provision

Long Service Award

Contract Completion Bonus

Future Benefits

Each provision is calculated independently.

10.3 Provision Types

The system supports unlimited provision types.

Examples

EOS Provision

Annual Leave Provision

Air Ticket Provision

Schooling Provision

Bonus Provision

Commission Provision

Contract Completion Provision

Future Provision Types

10.4 Provision Structure

Each Provision contains:

Provision Code

Provision Name

Category

Description

Calculation Method

Payroll Components Included

Frequency

Accounting Account

Currency

Labor Law Package

Company Policy

Status

Effective Date

Expiry Date

10.5 Monthly Provision

Every payroll period generates monthly provisions.

Example

Employee

Ahmed

Basic Salary

5,000

EOS Rate

8.33%

Monthly EOS Provision

416.50

This amount is recognized in Accounting only.

Employee does not receive this amount in Payroll.

10.6 EOS Provision

End of Service Provision depends on:

Labor Law

Contract Type

Employee Nationality

Company Policy

Years of Service

Eligible Salary Components

The Rule Engine determines the final formula.

10.7 Leave Provision

Unused Annual Leave creates Leave Provision.

Example

Annual Leave Balance

18 Days

Daily Salary

250

Leave Provision

18 × 250

=

4,500

This amount is recorded as Accounting Liability.

10.8 Air Ticket Provision

Certain contracts include Air Ticket entitlement.

Example

Air Ticket

Every 24 Months

Ticket Value

2,400

Monthly Provision

2,400

÷

24

=

100

Monthly Air Ticket Provision

100

10.9 Provision Frequency

Each Provision defines its calculation frequency.

Monthly

Quarterly

Semi Annual

Annual

Contract Completion

Manual

Rule Engine

10.10 Payroll Integration

Payroll generates:

Approved Salary

↓

Provision Engine

↓

Calculates Accruals

↓

Stores Accounting Transactions

↓

Accounting Module

Payroll never pays Provision amounts.

10.11 Accounting Integration

Each Provision generates:

Debit Account

Credit Account

Cost Center

Project

Cost Code

Accounting Period

Journal Reference

Posting Status

Journal Entries are configurable.

10.12 Provision Adjustment

Provision values may change.

Example

Salary Increase

Previous EOS Provision

400

New EOS Provision

480

Adjustment

80

The system automatically posts the adjustment.

10.13 Provision Reversal

When Provision is paid,

the accumulated liability must be reversed.

Example

EOS Paid

50,000

Accounting

Reverse EOS Provision

50,000

Create Payment Transaction

50,000

10.14 Final Settlement Integration

Final Settlement receives:

EOS

Leave Balance

Air Ticket

Bonus

Other Provisions

The Provision Engine transfers the accumulated liability automatically.

10.15 Business Scenario 1

Employee

Ahmed

Basic Salary

5,000

Housing

2,000

Transport

500

EOS Rule

Basic Salary Only

Provision

5,000 × 8.33%

=

416.50

10.16 Business Scenario 2

Employee

Mohammed

Annual Leave Balance

24 Days

Daily Salary

200

Leave Provision

24 × 200

=

4,800

10.17 Business Scenario 3

Employee

Fatima

Air Ticket

3,600

Every

36 Months

Monthly Provision

3,600

÷

36

=

100

10.18 Business Scenario 4

Employee resigned.

Provision Balance

EOS

42,000

Leave Provision

6,000

Air Ticket

2,000

Final Settlement

Automatically transfers all balances.

10.19 Validation

The Provision Engine validates:

Employee Active

Contract Exists

Payroll Completed

Rule Package Exists

Accounting Period Open

Payroll Components Valid

No Duplicate Provision

10.20 Audit Trail

Every Provision transaction records:

Provision Type

Calculation Formula

Payroll Period

Previous Value

New Value

Adjustment

Accounting Reference

User

Date

10.21 Performance Requirements

Support

70,000 Employees

Monthly Provision Calculation

Millions of Accounting Entries

Parallel Processing

Real-Time Provision Balance

10.22 Responsibilities

The Provision Engine is responsible for:

Provision Calculation

Provision Adjustment

Provision Reversal

Accounting Integration

Final Settlement Integration

Audit Trail

The Provision Engine never pays employees directly.

Payments are handled by the Payroll Engine or Final Settlement Engine.

End of Chapter 10

11.1 Introduction

The Additions & Deductions Engine manages all financial transactions that increase or decrease an employee's salary outside normal attendance calculations.

Unlike Payroll Components that originate from Contracts or Timesheets, Additions and Deductions may be:

- Manual
- Scheduled
- Recurring
- Rule Based
- Imported from external systems

The engine is fully integrated with:

- Payroll Engine
- Employee Management
- Accounting
- Workflow Engine
- Rule Engine
- Reporting Engine

11.2 Business Philosophy

Not every payroll transaction originates from attendance.

Some financial transactions occur independently.

Examples:

Performance Bonus

Commission

Schooling

Salary Advance

Loan Installment

Traffic Fine

Penalty

Medical Recovery

Mobile Bill

Accommodation Recovery

Fuel Recovery

Company Purchase

Future Adjustments

Each transaction is processed independently.

11.3 Transaction Categories

The engine supports unlimited categories.

Examples

Addition

Deduction

Loan

Advance

Bonus

Commission

Recovery

Allowance

Penalty

Manual Adjustment

Future Categories

11.4 Transaction Types

Every transaction contains:

Transaction Code

Transaction Name

Category

Description

Payroll Component

Calculation Method

Approval Required

Recurring

Schedule

Currency

Accounting Account

Status

Effective Date

Expiry Date

11.5 Additions

Examples

Performance Bonus

Attendance Bonus

Site Bonus

Safety Bonus

Retention Bonus

Annual Bonus

Referral Bonus

Manual Bonus

Adjustment

Reward

Commission

Future Additions

11.6 Deductions

Examples

Loan

Salary Advance

Insurance

Accommodation

Traffic Fine

Telephone

Fuel

Medical

Court Order

Penalty

Company Assets

Future Deductions

11.7 Recurring Transactions

Recurring transactions continue automatically until expiry.

Example

Mobile Allowance

150 QAR

Every Payroll

Start

January

End

December

The Payroll Engine automatically generates the transaction every month.

11.8 Scheduled Transactions

Some transactions occur at specific intervals.

Examples

Schooling

Every 6 Months

Air Ticket Recovery

Every 24 Months

Annual Bonus

Every December

Commission

Quarterly

The system executes them automatically.

11.9 One-Time Transactions

Examples

Joining Bonus

Retention Bonus

Salary Adjustment

Manual Recovery

Project Reward

Paid once only.

11.10 Loans

Loans are managed independently.

Loan Information

Loan Amount

Loan Balance

Installments

Monthly Installment

Interest (Optional)

Start Date

End Date

Approval

Payroll automatically deducts the monthly installment.

11.11 Salary Advance

Salary Advance differs from Loans.

Salary Advance

Amount

Approval

Recovery Date

Recovered automatically from Payroll.

11.12 Schooling

Schooling Allowance may be paid according to schedule.

Example

Employee

3 Children

Annual Schooling

18,000

Payment Frequency

Every 6 Months

Payroll generates

9,000

January

9,000

July

11.13 Bonus

Bonus calculation methods include:

Fixed Amount

Percentage

Performance Based

Manual

Company Profit Sharing

Rule Engine

11.14 Commission

Commission may depend on:

Sales

Projects

Invoices

Collections

Manual Entry

Rule Engine

11.15 Approval Workflow

Some transactions require approval.

Example

Employee

↓

Manager

↓

HR

↓

Finance

↓

Payroll

Approval levels are configurable.

11.16 Payroll Integration

Approved transactions are transferred automatically.

Payroll treats them as Payroll Components.

Gross Salary

-

Approved Additions

-

Approved Deductions

=

Net Salary

11.17 Accounting Integration

Every transaction generates accounting information.

Debit Account

Credit Account

Cost Center

Project

Accounting Period

Posting Status

Journal Number

11.18 Business Scenario 1

Employee

Ahmed

Performance Bonus

1,500

Payroll

Automatically adds

1,500

to Gross Salary.

11.19 Business Scenario 2

Employee

Mohammed

Loan

24,000

Installments

24

Monthly Recovery

1,000

Payroll automatically deducts

1,000

every month.

11.20 Business Scenario 3

Employee

Fatima

Schooling

24,000

Payment Frequency

Every 6 Months

Payroll

January

12,000

July

12,000

11.21 Business Scenario 4

Employee

Ali

Traffic Fine

750

Payroll

Deduct

750

Current Payroll

11.22 Validation

The engine validates:

Employee Exists

Payroll Period Open

Approval Completed

Transaction Active

Schedule Valid

Accounting Period Open

Currency Exists

No Duplicate Transaction

11.23 Audit Trail

Every transaction stores:

Reference Number

Payroll Period

Amount

Approval History

Accounting Reference

User

Date

Reason

Historical transactions shall never be deleted.

11.24 Performance Requirements

Support

70,000 Employees

Millions of Transactions

Bulk Import

Bulk Approval

Recurring Processing

Scheduled Processing

Real-Time Balance

11.25 Responsibilities

The Additions & Deductions Engine is responsible for:

Manual Transactions

Recurring Transactions

Scheduled Transactions

Loan Management

Salary Advance

Schooling

Bonus

Commission

Payroll Integration

Accounting Integration

Audit Trail

End of Chapter 11

Chapter 12

Final Settlement Engine

12.1 Introduction

The Final Settlement Engine is responsible for calculating all employee financial obligations at the end of employment.

The engine supports:

- Resignation
- Termination
- Contract Completion
- Retirement
- Death
- Transfer to another legal entity
- End of Fixed-Term Contract

The Final Settlement Engine consolidates all employee financial information into a single settlement transaction.

12.2 Business Philosophy

Final Settlement is not simply the employee's last salary.

It represents the complete financial relationship between the company and the employee.

The settlement determines:

- Amounts owed to the employee.
- Amounts owed by the employee.
- Company liabilities.
- Legal obligations.
- Accounting entries.
- Final payment.

The result must always be auditable and reproducible.

12.3 Settlement Inputs

The engine collects data from:

Employee Module

↓

Contract Module

↓

Payroll Engine

↓

Provision Engine

↓

Leave Engine

↓

Loan Module

↓

Asset Management

↓

Finance

↓

Accounting

↓

Rule Engine

↓

Labor Law Package

↓

Company Policy

12.4 Settlement Components

Typical Final Settlement includes:

Last Salary

Overtime

Undeclared Overtime (Approved Only)

Annual Leave Balance

Leave Encashment

End of Service (EOS)

Air Ticket

Bonus

Commission

Loans

Salary Advance

Penalties

Traffic Fines

Company Asset Recovery

Accommodation Recovery

Visa Cost Recovery

Other Adjustments

12.5 End of Service (EOS)

EOS shall be calculated according to:

Country Labor Law

Contract Type

Employee Category

Years of Service

Payroll Components Included

Company Policy

The Rule Engine performs the calculation.

No hard-coded formulas are permitted.

12.6 Leave Settlement

Unused leave shall be evaluated.

Each Leave Type defines:

Payable

Non-Payable

Expired

Carry Forward

Example

Annual Leave Balance

18 Days

Daily Rate

250

Leave Settlement

18 × 250

=

4,500

12.7 Air Ticket Settlement

Employees eligible for Air Ticket receive settlement according to:

Company Policy

Contract

Rule Package

Example

Ticket Value

3,000

Employee Eligible

Yes

Settlement

3,000

12.8 Loan Settlement

Outstanding loan balances are recovered.

Example

Loan

12,000

Paid

9,000

Remaining

3,000

Final Settlement automatically deducts

3,000

12.9 Company Asset Recovery

Before settlement approval, the system validates company assets.

Examples

Laptop

Mobile Phone

SIM Card

Vehicle

Uniform

Tools

Accommodation Key

ID Card

Access Card

If assets are not returned, recovery transactions are generated automatically.

12.10 Payroll Integration

The Payroll Engine transfers:

Last Salary

Approved Overtime

Manual Adjustments

Bonus

Commission

Current Payroll Balance

The Final Settlement Engine merges these values automatically.

12.11 Provision Integration

Provision Engine transfers:

EOS Provision

Leave Provision

Air Ticket Provision

Long Service Provision

The engine reverses provisions after payment.

12.12 Approval Workflow

Typical workflow:

Employee Exit Request

↓

Department Manager

↓

HR Department

↓

Finance

↓

Payroll

↓

Accounts

↓

General Manager (Optional)

↓

Settlement Approved

Workflow is fully configurable.

12.13 Settlement Payslip

The system generates a dedicated Settlement Payslip.

Contents include:

Employee Information

Employment Period

Reason for Exit

Salary Details

EOS

Leave Settlement

Air Ticket

Loans

Recoveries

Total Earnings

Total Deductions

Net Settlement

Approval History

12.14 Business Scenario 1

Employee

Ahmed

Service

6 Years

Basic Salary

5,000

EOS

Calculated according to Qatar Labor Law

Leave Balance

20 Days

Air Ticket

Eligible

Loan Balance

0

Settlement automatically includes:

EOS

Leave Encashment

Air Ticket

Last Salary

12.15 Business Scenario 2

Employee

Mohammed

Loan Balance

4,500

Leave Balance

10 Days

Air Ticket

Not Eligible

Final Settlement

Leave Encashment

Minus Loan Balance

Equals Net Settlement

12.16 Business Scenario 3

Employee

Fatima

Company Laptop

Not Returned

Laptop Value

2,000

Settlement automatically creates:

Company Asset Recovery

2,000

12.17 Business Scenario 4

Employee

Ali

Salary

8,000

Bonus

2,500

Approved OT

900

Loan

-1,500

Traffic Fine

-500

EOS

18,000

Leave Settlement

6,000

Net Final Settlement

Automatically calculated.

12.18 Validation

Before settlement approval:

Employee Status = Leaving

Payroll Closed

Assets Checked

Loans Updated

Leave Balance Calculated

Provision Updated

Accounting Period Open

Approval Completed

Settlement cannot be finalized until all validations succeed.

12.19 Accounting Integration

Settlement generates:

Journal Entries

Cost Allocation

Provision Reversal

Bank Payment

Accounting Reference

Payment Voucher

ERP Export

12.20 Audit Trail

Every Final Settlement stores:

Settlement Number

Employee

Reason

Calculation Version

Rule Package Version

Labor Law Version

Approval History

Accounting Reference

Payment Reference

User

Date

Historical settlements shall never be modified.

12.21 Performance Requirements

Support:

70,000+ Employees

Bulk Final Settlements

Parallel Processing

Real-Time Validation

Accounting Integration

ERP Integration

Bank Integration

12.22 Responsibilities

The Final Settlement Engine is responsible for:

EOS Calculation

Leave Settlement

Air Ticket Settlement

Provision Reversal

Loan Recovery

Asset Recovery

Accounting Integration

Settlement Payslip

Bank Payment

Audit Trail

The engine represents the final financial transaction in the employee lifecycle.

End of Chapter 12

Future Enhancements

The engine is designed to support future features without structural changes, including:

- Gratuity simulations before resignation.
- Multi-country labor law comparison.
- Automated visa cancellation workflow.
- Immigration document tracking.
- Electronic clearance forms.
- Digital signatures.
- Employee self-service settlement preview.
- AI-powered settlement validation and anomaly detection.

These enhancements can be enabled through configuration without changing the core architecture.

Chapter 13

Reporting & Analytics Engine

13.1 Introduction

The Reporting & Analytics Engine is responsible for transforming operational HR, Timesheet and Payroll data into business intelligence, operational reports, executive dashboards and external exports.

Unlike operational modules that process transactions, the Reporting Engine provides decision makers with accurate, real-time and historical information.

The Reporting Engine integrates with all modules of the system.

13.2 Objectives

The Reporting Engine shall:

- Provide operational reports.
- Provide payroll reports.
- Provide project cost reports.
- Provide executive dashboards.
- Support drill-down analysis.
- Support historical reporting.
- Support scheduled reports.
- Support custom reports.
- Export data to multiple formats.

13.3 Reporting Sources

Reports may retrieve data from:

Employee Module

↓

Contract Module

↓

Assignment Module

↓

Calendar

↓

Shift

↓

Timesheet

↓

Leave

↓

Overtime

↓

Payroll

↓

Provision

↓

Final Settlement

↓

Accounting

All reports shall use the same business rules as Payroll.

13.4 Report Categories

The system shall support unlimited report categories.

Examples

Operational Reports

Payroll Reports

Timesheet Reports

Leave Reports

Overtime Reports

Provision Reports

Accounting Reports

Project Reports

Cost Reports

Management Reports

Executive Dashboards

Government Reports

Audit Reports

Custom Reports

13.5 Employee Reports

Examples

Employee Master List

Employee Profile

Employee History

Employee Movement

Employee Assignment History

Employee Contract History

Employee Salary History

Employee Documents

Employee Expiry Report

Employee Age Analysis

Nationality Report

Visa Report

Medical Insurance Report

13.6 Timesheet Reports

Examples

Daily Attendance

Missing Attendance

Absent Report

Late Employees

Shift Utilization

Working Hours

Daily Hours Summary

Monthly Hours Summary

Crew Attendance

Cost Code Distribution

Project Hours

Attendance Exceptions

13.7 Payroll Reports

Examples

Payroll Register

Payroll Summary

Payroll Variance

Payroll Comparison

Payroll Audit

Payroll Components

Gross Salary Report

Net Salary Report

Payroll by Company

Payroll by Project

Payroll by Department

Payroll by Nationality

Payroll Cost Analysis

13.8 Leave Reports

Examples

Leave Balance

Leave Transactions

Leave Calendar

Leave Liability

Leave Forecast

Leave Approval Status

Leave Utilization

Unused Leave

Expired Leave

13.9 Overtime Reports

Examples

Declared Overtime

Undeclared Overtime

Holiday Overtime

Friday Overtime

Night Overtime

Approved Overtime

Rejected Overtime

Pending Overtime

Overtime Cost

13.10 Provision Reports

Examples

EOS Provision

Leave Provision

Air Ticket Provision

Monthly Provision

Provision Adjustment

Provision Reversal

Provision Liability

13.11 Project Reports

Examples

Project Labor Cost

Project Hours

Project Overtime

Project Headcount

Project Attendance

Project Cost Distribution

Cost Code Analysis

Labor Productivity

13.12 Financial Reports

Examples

Payroll Cost

Provision Liability

Department Cost

Project Cost

Cost Center Summary

Journal Summary

Payroll Forecast

Monthly Payroll Trend

13.13 Executive Dashboard

The system shall provide interactive dashboards.

Examples

Current Headcount

Employees on Leave

Employees on Site

Total Payroll

Payroll by Company

Payroll by Project

Payroll Trend

Attendance %

Absenteeism %

Overtime %

Leave Liability

EOS Liability

Labor Cost

Average Salary

Daily Workforce

13.14 Dashboard Widgets

Supported widgets include:

KPI Cards

Tables

Charts

Bar Charts

Line Charts

Pie Charts

Heat Maps

Trend Analysis

Geographical Maps (Optional)

All widgets shall support filtering.

13.15 Report Filters

Every report shall support dynamic filtering.

Examples

Company

Business Unit

Project

Department

Section

Employee

Nationality

Employee Type

Payroll Period

Calendar Year

Shift

Time Type

Leave Type

Overtime Type

Status

Cost Code

Cost Center

13.16 Export Formats

Reports may be exported as:

Excel (.xlsx)

CSV

PDF

Word

JSON

XML

REST API

The export format shall be configurable.

13.17 Scheduled Reports

The system supports automatic report generation.

Examples

Daily

Weekly

Monthly

Quarterly

Yearly

Reports may be delivered through:

Email

Shared Folder

API

Cloud Storage

13.18 Business Scenario 1

Project Manager

Requests

Monthly Labor Cost

Project

Doha Metro

Period

July 2026

System generates:

Headcount

Working Hours

Overtime

Payroll Cost

Cost by Cost Code

Cost by Department

13.19 Business Scenario 2

Payroll Manager

Requests

Payroll Comparison

June

vs

July

System displays:

Salary Difference

Bonus Difference

Loan Difference

Net Payroll Difference

Employee Count Difference

13.20 Business Scenario 3

HR Manager

Requests

Leave Balance Report

System displays:

Employee

Leave Type

Available Balance

Used Balance

Pending Requests

Forecast Balance

13.21 Business Scenario 4

CEO Dashboard

Displays:

Today's Headcount

Employees on Leave

Employees Working

Total Payroll

Total Overtime

Total Labor Cost

EOS Liability

Leave Liability

Real-time KPIs

13.22 Report Security

Reports shall respect Role-Based Access Control (RBAC).

Examples

HR Officer

Employee Reports

Payroll Officer

Payroll Reports

Project Manager

Project Reports Only

Finance

Financial Reports

CEO

All Reports

Users shall never access unauthorized information.

13.23 Performance Requirements

The Reporting Engine shall support:

70,000+ Employees

Millions of Transactions

Historical Reports

Real-Time Dashboards

Background Report Generation

Parallel Report Processing

Export Large Excel Files

13.24 Audit

Every generated report stores:

Report Name

User

Filters

Generation Time

Export Format

Execution Duration

Report Version

Audit records support compliance and troubleshooting.

13.25 Responsibilities

The Reporting & Analytics Engine is responsible for:

Operational Reporting

Payroll Reporting

Leave Reporting

Overtime Reporting

Provision Reporting

Accounting Reporting

Executive Dashboards

Business Intelligence

Scheduled Reports

Export Services

Historical Analysis

The Reporting Engine never modifies operational data.

It is strictly read-only.

End of Chapter 13

Future Enhancements

The Reporting Engine is designed to support future capabilities without architectural changes:

- AI-powered insights and anomaly detection.
- Predictive payroll forecasting.
- Leave trend prediction.
- Workforce planning analytics.
- Interactive drill-through dashboards.
- Power BI integration.

- Tableau integration.
- Microsoft Excel Live connection.
- REST Analytics API.
- Custom report designer for end users.

Chapter 14

Integration & Export Engine

14.1 Introduction

The Integration & Export Engine is responsible for exchanging information between the Workforce Management & Payroll System and external applications.

The system shall support bidirectional integration.

Inbound integrations import data into the system.

Outbound integrations publish validated business data to external systems.

The Integration Engine shall support both real-time APIs and scheduled batch processing.

14.2 Objectives

The Integration Engine shall:

- Eliminate manual data entry.
- Integrate with ERP systems.
- Integrate with Finance systems.
- Integrate with Attendance devices.
- Integrate with Banks.
- Integrate with Government portals.
- Support Excel Import/Export.
- Support configurable interfaces.
- Support future systems without software modification.

14.3 Integration Architecture

External System

↓

Integration Layer

↓

Validation Engine

↓

Business Rules

↓

Core Modules

↓

Audit Log

↓

Response

Every integration shall pass through validation before reaching business modules.

14.4 Integration Types

Supported integration methods:

REST API

SOAP

CSV

Excel

XML

JSON

SFTP

Database Views

Database Procedures

Message Queue

Webhook

Future Connectors

14.5 Import Sources

The system shall support importing data from:

Attendance Machines

Biometric Devices

Excel Files

CSV Files

Payroll Legacy Systems

ERP Systems

Finance Systems

Recruitment Systems

Employee Master Systems

Government Systems

Third-Party Applications

14.6 Export Targets

The system shall export data to:

ERP

Finance

SAP

Oracle ERP

Microsoft Dynamics

Banks

WPS

Government Portals

Business Intelligence

Power BI

Tableau

External Payroll Systems

Custom Applications

14.7 Excel Import

Excel import shall support:

Employee Master

Employee Contracts

Employee Assignment

Timesheets

Leave

Overtime

Payroll Adjustments

Additions

Deductions

Projects

Cost Codes

Payroll Components

Shifts

Calendars

Rule Packages

The system validates all imported data before processing.

14.8 Excel Export

Excel exports shall support:

Employee Reports

Payroll Reports

Timesheet Reports

Cost Reports

Project Reports

Leave Reports

Provision Reports

Accounting Reports

Audit Reports

Executive Reports

Exports shall preserve formatting and support large datasets.

14.9 Attendance Integration

Attendance devices provide raw attendance events.

Example

Badge Number

Date

Check In

Check Out

Device

Location

These records are validated and transferred to the Timesheet Engine.

Attendance devices never update Payroll directly.

14.10 Payroll Export

Payroll exports include:

Employee Number

Employee Name

Payroll Period

Gross Salary

Net Salary

Bank Information

Payroll Components

Cost Allocation

Currency

Export Status

Reference Number

14.11 WPS Export

The system shall generate Wage Protection System files according to country requirements.

Supported countries include:

Qatar

United Arab Emirates

Oman

Saudi Arabia

Future countries may be added through configuration.

Validation shall occur before file generation.

14.12 Bank Integration

Bank export supports:

Bank Code

Branch Code

IBAN

Account Number

Employee Name

Currency

Transfer Amount

Reference Number

Banks shall be maintained as Master Data.

Each employee references one active bank account.

14.13 ERP Integration

The system shall export:

Payroll Journal Entries

Provision Entries

Cost Allocation

Project Costs

Department Costs

Accounting Period

Posting Status

ERP Reference

Integration shall support multiple ERP vendors.

14.14 API Integration

REST APIs shall support:

Employee API

Contract API

Assignment API

Timesheet API

Leave API

Payroll API

Provision API

Reporting API

Approval API

Notification API

Every API shall be versioned.

14.15 Mapping Engine

The system shall include a configurable Mapping Engine.

Instead of hardcoding export formats, administrators define mappings.

Example

Employee Number

↓

Column A

Employee Name

↓

Column B

Basic Salary

↓

Column H

Net Salary

↓

Column M

IBAN

↓

Column R

Changing customer formats shall require mapping changes only.

14.16 Scheduled Interfaces

Interfaces may execute:

Hourly

Daily

Weekly

Monthly

On Demand

Event Driven

Each interface stores:

Execution Time

Status

Records Processed

Errors

Duration

14.17 Error Handling

Failed integrations shall never lose data.

Each failed transaction stores:

Interface Name

Transaction Reference

Payload

Error Description

Retry Count

Status

Resolved By

Resolution Date

Automatic retry shall be supported.

14.18 Business Scenario 1

Customer requests payroll export.

Format

Excel

Period

July 2026

System generates:

Payroll Register

Employee Details

Net Salary

IBAN

Payroll Components

Export completed successfully.

14.19 Business Scenario 2

Attendance device uploads

125,000 attendance records.

Validation identifies:

120 valid records

5 invalid badge numbers

Valid records imported.

Invalid records remain in Exception Queue.

14.20 Business Scenario 3

Finance requests Journal Entries.

Payroll completed.

System exports:

Debit Account

Credit Account

Amount

Cost Center

Project

Accounting Period

Journal Number

14.21 Business Scenario 4

Customer changes ERP format.

Administrator updates Mapping Engine.

No software modification.

No database modification.

Next export follows the new layout automatically.

14.22 Security

All integrations shall support:

Authentication

Authorization

Encryption

Digital Signatures (Optional)

Audit Logging

API Keys

OAuth2 / JWT (where applicable)

Transport Layer Security (TLS)

14.23 Performance Requirements

Support:

70,000+ Employees

Millions of Export Records

Large Excel Files

Parallel Processing

Background Processing

Scheduled Interfaces

Real-Time APIs

Automatic Retry

14.24 Audit Trail

Every integration stores:

Interface Name

Direction

Execution Time

Duration

User

Records Processed

Success Count

Failure Count

Reference Number

Payload Hash

Audit information shall remain permanently available.

14.25 Responsibilities

The Integration & Export Engine is responsible for:

Import Services

Export Services

API Management

File Generation

Bank Interfaces

ERP Interfaces

Attendance Integration

Government Interfaces

Mapping Engine

Audit

Retry Management

The Integration Engine shall never perform Payroll calculations.

Its responsibility is secure and reliable data exchange only.

Future Enhancements

Future enhancements include:

- GraphQL API support.
- Event Streaming (Kafka/RabbitMQ).
- AI-assisted mapping recommendations.
- Cloud storage connectors.
- SharePoint integration.
- Google Drive / OneDrive export.
- S3-compatible object storage.
- Low-code integration designer.
- Webhook subscriptions.
- Enterprise Service Bus (ESB) integration.

End of Chapter 14

Chapter 15

Business Rule Engine

15.1 Introduction

The Business Rule Engine is the decision-making core of the Workforce Management & Payroll System.

Unlike traditional payroll systems where business logic is hardcoded inside application code, this system stores business rules as configurable metadata.

The Rule Engine evaluates every payroll transaction using configurable rules without requiring software modification.

Every module communicates with the Rule Engine before executing business logic.

15.2 Philosophy

The system shall never hardcode:

Labor Laws

Company Policies

Payroll Calculations

Leave Rules

Overtime Rules

Provision Rules

Instead,

Business Rules shall be configurable.

Changing a rule shall never require software deployment.

15.3 Rule Engine Scope

The Rule Engine is responsible for:

Payroll Rules

Leave Rules

Shift Rules

Calendar Rules

Time Type Rules

Provision Rules

EOS Rules

Overtime Rules

Approval Rules

Validation Rules

Notification Rules

Future Business Rules

15.4 Rule Hierarchy

The Rule Engine evaluates rules according to the following priority.

Global Rules

↓

Country Rules

↓

Company Rules

↓

Business Unit Rules

↓

Project Rules

↓

Employee Category Rules

↓

Contract Rules

↓

Employee Rules

The lowest level overrides the higher level.

15.5 Rule Package

Rules are grouped into Rule Packages.

Examples

Qatar Labor Law

UAE Labor Law

Saudi Labor Law

Company Policy

Project Policy

Client Policy

Every employee references one active Rule Package.

15.6 Rule Definition

Every Rule contains:

Rule Code

Rule Name

Rule Category

Description

Rule Type

Priority

Effective Date

Expiry Date

Version

Status

Created By

Approved By

15.7 Rule Types

Supported Rule Types include:

Calculation Rule

Validation Rule

Approval Rule

Notification Rule

Payroll Rule

Leave Rule

Shift Rule

Provision Rule

Accounting Rule

Future Rule Types

15.8 Rule Conditions

Every Rule consists of:

Condition

Operator

Value

Result

Example

IF

Time Type

=

Annual Leave

THEN

Basic Salary

=

Paid

Housing

=

Paid

Transport

=

Paid

Attendance Bonus

=

Not Paid

15.9 Payroll Rule Example

IF

Employee Type

=

Monthly

AND

Worked Days

0

THEN

Pay Basic Salary

Otherwise

No Salary

15.10 Leave Rule Example

IF

Leave Type

=

Annual Leave

THEN

Deduct Leave Balance

Pay Basic Salary

Include EOS

Generate Leave Provision

15.11 Overtime Rule Example

IF

Worked Hours

Shift Hours

AND

Declared OT Available

THEN

Generate Declared OT

ELSE

Generate Undeclared OT

15.12 Holiday Rule Example

IF

Working Day

=

Public Holiday

THEN

Holiday Rate

=

200%

Generate Holiday Overtime

15.13 Friday Rule Example

IF

Working Day

=

Friday

THEN

Friday Rate

=

150%

15.14 Shift Rule Example

IF

Shift

=

8+2

Worked

=

11

THEN

Normal

8

Declared OT

2

Undeclared OT

1

15.15 Leave Accrual Rule

IF

Employee Completed

1 Month

THEN

Accrue

2.5 Days

Annual Leave

15.16 EOS Rule

IF

Service

5 Years

THEN

EOS Formula

Version 2

Otherwise

EOS Formula

Version 1

15.17 Rule Versioning

Business Rules shall never be modified directly.

Instead

Version 1

↓

Version 2

↓

Version 3

Historical Payroll always references the rule version active during calculation.

15.18 Rule Testing

Administrators may simulate Rules before activation.

Example

Employee

Ahmed

Payroll Period

July

Rule Package

Qatar 2026

Simulation Result

Displayed without updating Payroll.

15.19 Rule Approval

Changes to Rule Packages require approval.

Example

Payroll Administrator

↓

HR Manager

↓

Finance

↓

System Administrator

↓

Publish

15.20 Rule Activation

Rules become active according to:

Effective Date

Company

Project

Employee Category

Payroll Period

No application restart is required.

15.21 Business Scenario 1

Qatar Labor Law changes Annual Leave calculation.

Administrator creates:

Annual Leave Rule

Version 2

Effective

01-Jan-2027

Payroll

2026

continues using

Version 1

Payroll

2027

uses

Version 2

15.22 Business Scenario 2

Company A pays Housing during Sick Leave.

Company B does not.

Both companies use the same software.

Different Rule Packages.

No code modification.

15.23 Business Scenario 3

Project ABC

Approves only

2 OT Hours

Project XYZ

Approves

4 OT Hours

The Rule Engine applies different Project Policies automatically.

15.24 Validation

Every Rule is validated before activation.

Checks include:

Syntax

Circular Dependency

Duplicate Rule

Priority Conflict

Effective Date Conflict

Missing References

Invalid rules cannot be activated.

15.25 Performance Requirements

Support:

Thousands of Rules

Multiple Versions

Real-Time Evaluation

Parallel Processing

Simulation Mode

Instant Activation

15.26 Audit Trail

Every Rule stores:

Version

Previous Value

New Value

User

Approval History

Activation Date

Affected Modules

Historical rules are never deleted.

15.27 Responsibilities

The Business Rule Engine is responsible for:

Payroll Rules

Leave Rules

Overtime Rules

Provision Rules

Validation Rules

Approval Rules

Notification Rules

Rule Versioning

Simulation

Audit

Every business decision inside the system shall be driven by the Rule Engine.

No business logic shall be hardcoded inside application code.

Future Enhancements

The Rule Engine is designed to support:

- Visual Rule Builder.
- Drag-and-drop rule designer.
- AI-generated rule recommendations.

- Natural language rule creation.
- Decision tables (DMN).
- BPMN workflow integration.
- Machine learning validation.
- Rule impact analysis.
- Rule dependency graph.
- Multi-language rule descriptions.

End of Chapter 15

Chapter 16

System Architecture & Technical Design

16.1 Introduction

The Workforce Management & Payroll System is designed as an Enterprise-grade application capable of supporting large organizations with more than 70,000 employees across multiple companies, projects and countries.

The architecture follows modern distributed software engineering principles to ensure scalability, maintainability, security and performance.

The system shall support future expansion without major architectural changes.

16.2 Architecture Principles

The system architecture shall follow the following principles:

- Modular Design
- Domain Driven Design (DDD)
- Separation of Concerns
- Clean Architecture
- SOLID Principles
- Configuration Driven Business Logic

- API First Design
- Event Driven Integration
- High Availability
- Horizontal Scalability

16.3 High-Level Architecture

```
                React Web Application
                         │
                React Mobile (Future)
                         │
               REST API / HTTPS / JWT
                         │
                 Spring Boot Backend
                         │
┌──────────────────────────────────────────────────────────┐
│ Employee Module                                          │
│ Contract Module                                          │
│ Assignment Module                                        │
│ Calendar Module                                          │
│ Shift Module                                             │
│ Timesheet Module                                         │
│ Leave Module                                             │
│ Overtime Module                                          │
│ Payroll Module                                           │
│ Provision Module                                         │
│ Final Settlement Module                                  │
│ Reporting Module                                         │
│ Integration Module                                       │
│ Rule Engine                                              │
└──────────────────────────────────────────────────────────┘
                         │
             PostgreSQL Database Cluster
                         │
      Redis │ RabbitMQ │ MinIO │ Backup Storage
```

16.4 Technology Stack

Backend

Java 21

Spring Boot

Spring Security

Spring Data JPA

Hibernate

MapStruct

Lombok

Flyway

Maven

Frontend

React

TypeScript

Material UI

React Query

React Router

Axios

Formik

Yup

AG Grid

Database

PostgreSQL

UUID Primary Keys

Partitioned Tables

Materialized Views

GIN Indexes

JSONB Support

Infrastructure

Docker

Docker Compose

Kubernetes (Future)

Nginx

Redis

RabbitMQ

MinIO

Prometheus

Grafana

ELK Stack

16.5 Modular Architecture

Every business domain shall be implemented as an independent module.

Core Modules

Employee

Contract

Assignment

Calendar

Shift

Time Type

Payroll Components

Operational Modules

Timesheet

Leave

Overtime

Payroll

Provision

Settlement

System Modules

Workflow

Approval

Notification

Audit

Attachment

Reporting

Integration

Rule Engine

16.6 Backend Layers

Presentation Layer

↓

REST Controllers

↓

Application Services

↓

Business Services

↓

Rule Engine

↓

Repositories

↓

Database

Business logic shall never exist inside Controllers.

16.7 Frontend Architecture

The React application shall consist of:

Pages

↓

Layouts

↓

Components

↓

Forms

↓

Tables

↓

Dialogs

↓

API Services

↓

State Management

↓

Theme

Reusable components shall be used throughout the application.

16.8 Security Architecture

Authentication

JWT

Refresh Tokens

Role Based Access Control (RBAC)

Password Encryption

HTTPS

Audit Logging

Session Timeout

Optional Multi-Factor Authentication (MFA)

16.9 Background Processing

Long-running tasks shall execute asynchronously.

Examples

Payroll Run

Large Excel Import

Excel Export

Report Generation

Bank File Generation

Provision Calculation

Email Notifications

Background jobs shall not block user requests.

16.10 File Storage

The system stores:

Employee Documents

Contracts

Medical Certificates

Leave Attachments

Payroll Attachments

Settlement Documents

Reports

Generated Files

Files shall be stored in object storage (MinIO/S3 compatible).

16.11 Caching

Redis shall cache:

Master Data

Rule Packages

Calendars

Shift Definitions

Lookup Tables

Session Data

Frequently Requested Reports

Cache invalidation shall occur automatically when master data changes.

16.12 Messaging

RabbitMQ shall be used for:

Payroll Events

Notification Events

Integration Events

Report Generation

Background Processing

Future Event Streaming

16.13 Logging

Every request shall generate:

Request ID

User

Module

API

Execution Time

Response Status

Error Details

Correlation ID

Logs shall support troubleshooting and auditing.

16.14 Monitoring

System monitoring shall include:

CPU Usage

Memory Usage

Database Performance

API Response Time

Queue Status

Background Jobs

Failed Integrations

Disk Usage

Application Health

Prometheus and Grafana shall be used.

16.15 High Availability

The architecture supports:

Multiple Application Servers

Load Balancer

Database Replication

Automatic Failover

Rolling Deployment

Zero Downtime Updates

16.16 Performance Targets

Support:

70,000+ Employees

500 Concurrent Users

Millions of Payroll Transactions

Payroll < 10 Minutes

Report Generation < 30 Seconds

API Response < 500 ms

Excel Export > 1 Million Rows

16.17 Deployment

Development Environment

Testing Environment

UAT Environment

Production Environment

Each environment shall have independent configuration.

16.18 Configuration Management

Configuration shall include:

Application Settings

Database Connections

Rule Packages

Email Settings

Notification Settings

File Storage

Authentication

Integration Endpoints

No environment-specific values shall be hardcoded.

16.19 Disaster Recovery

The system shall support:

Daily Backup

Point-in-Time Recovery

Database Replication

File Backup

Recovery Testing

Backup Encryption

16.20 Future Scalability

The architecture is designed to support future modules without redesign.

Examples

Recruitment

Performance Management

Training

Medical Insurance

Asset Management

Visitor Management

Fleet Management

AI Workforce Analytics

16.21 Responsibilities

The Technical Architecture is responsible for:

Application Structure

Technology Stack

Scalability

Performance

Security

Availability

Monitoring

Maintainability

Future Expansion

The architecture separates technical implementation from business rules, allowing continuous evolution without affecting business functionality.

End of Chapter 16

Chapter 17

System Data Architecture & Domain Model

17.1 Introduction

The purpose of the Data Architecture is to define how information is organized inside the Workforce Management & Payroll System.

Unlike traditional HRMS systems where database tables dictate business logic, this system follows a Domain-Driven Design (DDD) approach.

Business Domains are defined first.

Database implementation follows the business domains.

17.2 Design Philosophy

The database shall never drive business logic.

Instead,

Business Rules

↓

Domain Model

↓

Database Design

↓

Application Code

This approach allows future business changes without redesigning the database.

17.3 Core Domains

The entire system is divided into Core Domains.

Employee Domain

Contract Domain

Assignment Domain

Calendar Domain

Shift Domain

Payroll Domain

Timesheet Domain

Leave Domain

Overtime Domain

Provision Domain

Settlement Domain

Workflow Domain

Reporting Domain

Integration Domain

Rule Engine Domain

Notification Domain

Audit Domain

Master Data Domain

Each domain owns its own business objects.

17.4 Employee Domain

Responsible for employee identity.

Contains:

Employee

Nationality

Religion

Marital Status

Dependents

Documents

Bank Accounts

Emergency Contacts

Medical Information

Employment Status

The Employee Domain never stores payroll transactions.

17.5 Contract Domain

Responsible for employment contracts.

Contains:

Employment Contract

Salary Package

Payroll Frequency

Labor Law

Company Policy

Effective Dates

Contract Amendments

Multiple contracts may exist during employee lifetime.

17.6 Assignment Domain

Responsible for employee deployment.

Contains:

Company

Business Unit

Project

Department

Section

Supervisor

Shift Assignment

Cost Center

Cost Code

Employees may have multiple assignments over time.

17.7 Calendar Domain

Responsible for:

Years

Periods

Months

Public Holidays

Working Days

Payroll Status

Calendar Versions

Multiple calendars may coexist.

17.8 Shift Domain

Responsible for:

Shift Definitions

Working Hours

Breaks

Night Hours

Weekly Off

Holiday Rules

Declared OT

Maximum Hours

Shift Rotation

Employees may move between shifts.

17.9 Timesheet Domain

Responsible for operational attendance.

Contains:

Timesheet Header

Daily Records

Work Segments

Time Types

Cost Allocation

Attendance Source

Approval

Validation

The Timesheet Domain never stores salary.

17.10 Payroll Domain

Responsible for:

Payroll Run

Payroll Transactions

Payroll Components

Gross Salary

Net Salary

Taxes

Currency

Bank Payment

Payslip

Payroll Lock

Payroll Version

17.11 Leave Domain

Contains:

Leave Requests

Leave Balance

Leave Accrual

Leave Approval

Leave Encashment

Leave Settlement

Leave History

17.12 Overtime Domain

Contains:

Declared OT

Undeclared OT

Holiday OT

Friday OT

Night OT

Approvals

Payment Status

17.13 Provision Domain

Contains:

EOS Provision

Leave Provision

Air Ticket Provision

Provision Adjustment

Provision Reversal

Accounting Posting

17.14 Financial Transaction Domain

Contains all financial movements that are not generated directly from Timesheets.

Examples

Bonus

Loan

Commission

Schooling

Salary Advance

Penalty

Traffic Fine

Manual Adjustment

Recurring Payments

Scheduled Payments

Payroll consumes these transactions.

17.15 Settlement Domain

Responsible for:

EOS

Leave Settlement

Provision Reversal

Loans

Recoveries

Asset Recovery

Final Payslip

Settlement Payment

17.16 Workflow Domain

Responsible for:

Approval Templates

Approval Steps

Delegation

Escalation

Approval History

Dynamic Approval Routing

Every business module may use Workflow.

17.17 Notification Domain

Responsible for:

Email

SMS

Push Notification

System Notification

Scheduled Reminder

Expiry Alerts

Notification History

17.18 Reporting Domain

Responsible for:

Operational Reports

Analytical Reports

Dashboards

Scheduled Reports

BI Integration

Exports

The Reporting Domain is Read Only.

17.19 Integration Domain

Responsible for:

REST APIs

Excel Import

Excel Export

Bank Integration

ERP Integration

Attendance Integration

Government Interfaces

Mapping Engine

17.20 Rule Engine Domain

Responsible for all configurable business rules.

Payroll Rules

Leave Rules

Shift Rules

Time Type Rules

Validation Rules

Approval Rules

Notification Rules

No module contains hardcoded business logic.

17.21 Audit Domain

Every transaction generates audit information.

Examples

Created By

Modified By

Approval History

Rule Version

Before Value

After Value

Timestamp

Audit records are immutable.

17.22 Master Data Domain

Contains all reusable reference information.

Examples

Companies

Projects

Departments

Sections

Banks

Currencies

Countries

Nationalities

Cost Centers

Cost Codes

Payroll Components

Leave Types

Time Types

Shifts

Calendars

Rule Packages

Master Data is shared across all modules.

17.23 Domain Relationships

Employee

↓

Contract

↓

Assignment

↓

Timesheet

↓

Payroll

↓

Provision

↓

Settlement

Each domain communicates only through business services.

Direct database dependencies shall be minimized.

17.24 Scalability

Every domain shall support:

Multi Company

Multi Country

Multi Currency

Multi Language

Unlimited Projects

Unlimited Employees

Unlimited Payroll Periods

Unlimited Historical Data

17.25 Responsibilities

The Domain Model is responsible for:

Business Separation

Module Independence

Future Expansion

Performance

Maintainability

Database Simplicity

The Domain Model is the blueprint for the physical database that will be designed in the next chapters.

End of Chapter 17

Chapter 18

Database Design Principles

18.1 Introduction

This chapter defines the database design principles for the Enterprise Workforce Management & Payroll System.

The database is designed using PostgreSQL and follows modern enterprise architecture principles.

The database is expected to support:

- 70,000+ Employees
- Multiple Companies

- Multiple Countries
- Multiple Payroll Cycles
- Unlimited Historical Data
- Millions of Monthly Transactions

The primary objective is to create a scalable, maintainable and high-performance database.

18.2 Design Philosophy

The database is designed according to the following principles.

Business First

↓

Domain Driven Design

↓

Normalized Database

↓

Performance Optimization

↓

Scalability

The database shall never contain business logic.

Business Rules belong exclusively to the Rule Engine.

18.3 Naming Convention

Tables

Singular

Examples

Employee

Contract

Project

PayrollRun

LeaveRequest

PayrollTransaction

Columns

camelCase

Examples

employeeId

employeeNumber

companyId

projectId

createdAt

updatedAt

Primary Keys

UUID

Foreign Keys

Reference parent UUID.

18.4 UUID Strategy

Every business table shall use UUID as Primary Key.

Advantages

Global uniqueness

Distributed systems support

No dependency on database sequences

Safe replication

Microservice ready

Example

Employee

employeeId UUID

Contract

contractId UUID

Project

projectId UUID

18.5 Audit Columns

Every transactional table contains:

createdAt

createdBy

updatedAt

updatedBy

version

isDeleted

deletedAt

deletedBy

Audit information shall never be optional.

18.6 Soft Delete

Business records are never physically deleted.

Instead,

isDeleted = true

Historical records remain available for:

Audit

Payroll Reproduction

Legal Compliance

Reporting

18.7 Master Data

Master Data tables contain:

Company

BusinessUnit

Department

Section

Project

Bank

Currency

Country

Nationality

Religion

CostCenter

CostCode

PayrollComponent

LeaveType

TimeType

Shift

Calendar

RulePackage

Master Data changes infrequently.

18.8 Transaction Tables

Transaction tables include:

Timesheet

Payroll

Leave

Overtime

Provision

Settlement

Approval

Notification

FinancialTransaction

Audit

Transaction tables grow continuously.

18.9 Reference Tables

Reference tables include:

Gender

Marital Status

Education

Employment Type

Employment Status

Payroll Frequency

Approval Status

Workflow Status

Document Type

Reference tables rarely change.

18.10 Table Classification

The system classifies tables into four categories.

Master

Reference

Transaction

History

Each category follows different indexing and storage strategies.

18.11 History Tables

Historical information shall never overwrite current records.

Examples

Contract History

Salary History

Assignment History

Shift History

Bank History

Document History

Payroll History

Leave History

Each change creates a new historical record.

18.12 Effective Dating

Business entities support effective dates.

Every record contains:

effectiveFrom

effectiveTo

This allows:

Future Changes

Historical Reporting

Retroactive Payroll

Contract Amendments

18.13 Versioning

Business entities support versioning.

Example

Contract

Version 1

↓

Version 2

↓

Version 3

Payroll always references the version used during calculation.

18.14 Database Relationships

Relationships shall use Foreign Keys.

Examples

Employee

↓

Contract

↓

Assignment

↓

Payroll

↓

Payslip

Many-to-many relationships shall use junction tables.

18.15 Partition Strategy

Large tables shall be partitioned.

Examples

Timesheet

Partition

Payroll Period

PayrollTransaction

Partition

Payroll Period

Attendance

Partition

Month

Audit

Partition

Year

Partitioning improves performance and maintenance.

18.16 Index Strategy

Indexes shall be created on:

Primary Keys

Foreign Keys

Employee Number

Payroll Period

Company

Project

Status

Approval Status

Date Fields

Composite indexes shall be used where appropriate.

18.17 Constraints

The database shall enforce:

Primary Keys

Foreign Keys

Unique Constraints

Check Constraints

Not Null Constraints

Business validations remain inside the application layer.

18.18 Database Performance

Performance techniques include:

Partitioning

Materialized Views

GIN Indexes

Connection Pooling

Prepared Statements

Bulk Inserts

Batch Updates

Read Optimization

18.19 Backup Strategy

The system supports:

Daily Backup

Incremental Backup

Point-in-Time Recovery

Replication

Archive

Backup Encryption

Regular Recovery Testing

18.20 Data Retention

The system stores:

Payroll

Permanent

Employee History

Permanent

Audit

Permanent

Notifications

Configurable

Temporary Imports

Configurable

Retention policies shall comply with company policy and legal requirements.

18.21 Security

Sensitive information shall be protected.

Examples

Salary

Bank Account

IBAN

National ID

Passport

Medical Information

Security includes:

Encryption

Role-Based Access

Audit

Database Security

TLS Connections

18.22 Scalability

The database supports:

70,000+ Employees

Unlimited Projects

Unlimited Payroll Runs

Unlimited Historical Data

Millions of Transactions

Multi Company

Multi Country

Multi Currency

18.23 Responsibilities

The Database Design is responsible for:

Data Integrity

Performance

Scalability

Historical Accuracy

Security

Maintainability

The physical database shall always reflect the Business Domain Model defined in previous chapters.

End of Chapter 18

Chapter 19

Database Schema Design

19.1 Introduction

This chapter defines the logical database schema for the Enterprise Workforce Management & Payroll System.

The objective is to create a database capable of supporting enterprise organizations with more than 70,000 employees while maintaining high performance, scalability and data integrity.

The schema is designed specifically for PostgreSQL.

19.2 Database Layers

The database is divided into logical schemas.

Master Data

↓

Core HR

↓

Operations

↓

Payroll

↓

Accounting

↓

Workflow

↓

Security

↓

Audit

↓

Reporting

Each layer has a clear responsibility.

19.3 Master Data Schema

Contains static reference information.

Examples

Company

Business Unit

Department

Section

Project

Location

Country

Nationality

Currency

Bank

Cost Center

Cost Code

Calendar

Shift

Payroll Component

Time Type

Leave Type

Rule Package

Master Data changes infrequently.

19.4 Core HR Schema

Contains employee master information.

Entities include

Employee

Employee Address

Employee Contact

Employee Bank

Employee Documents

Employee Dependents

Employee Emergency Contact

Employee Education

Employee Experience

Employee Skills

Employee Medical

Employee Identity

19.5 Contract Schema

Contains employment contracts.

Entities

Contract

Salary Package

Contract Amendment

Payroll Frequency

Labor Law Assignment

Company Policy Assignment

Contract History

19.6 Assignment Schema

Responsible for employee deployment.

Entities

Assignment

Company Assignment

Project Assignment

Department Assignment

Supervisor Assignment

Shift Assignment

Cost Center Assignment

Cost Code Assignment

Assignment History

19.7 Calendar Schema

Entities

Calendar

Calendar Year

Payroll Period

Holiday

Holiday Group

Calendar Version

Period Status

19.8 Shift Schema

Entities

Shift

Shift Hours

Break

Rotation

Shift Rule

Shift History

19.9 Timesheet Schema

Entities

Timesheet Header

Timesheet Detail

Work Segment

Cost Allocation

Attendance Event

Approval

Correction

Exception

Daily Summary

19.10 Leave Schema

Entities

Leave Request

Leave Balance

Leave Accrual

Leave Approval

Leave Attachment

Leave Settlement

Leave History

19.11 Overtime Schema

Entities

Overtime Request

Declared Overtime

Undeclared Overtime

Holiday Overtime

Night Overtime

Approval

Payment

History

19.12 Payroll Schema

Entities

Payroll Run

Payroll Transaction

Payroll Component

Payroll Detail

Payslip

Payroll Bank Transfer

Payroll Audit

Payroll History

19.13 Financial Schema

Entities

Employee Financial Transaction

Loan

Salary Advance

Bonus

Commission

Penalty

Schooling

Manual Adjustment

Recurring Transaction

Scheduled Transaction

19.14 Provision Schema

Entities

Provision Transaction

EOS Provision

Leave Provision

Air Ticket Provision

Provision Adjustment

Provision Reversal

19.15 Settlement Schema

Entities

Settlement

Settlement Detail

Settlement Approval

Settlement Payment

Settlement Recovery

Settlement History

19.16 Workflow Schema

Entities

Workflow Template

Workflow Step

Approval

Approval History

Delegation

Escalation

Workflow Log

19.17 Notification Schema

Entities

Notification

Notification Queue

Email

SMS

Push Notification

Reminder

Notification History

19.18 Security Schema

Entities

User

Role

Permission

Role Permission

User Role

Session

Login History

Password History

API Token

19.19 Audit Schema

Entities

Audit Log

Entity History

Rule History

Payroll Audit

API Audit

Login Audit

Change Tracking

19.20 Reporting Schema

Contains read-only objects.

Materialized Views

Summary Tables

Dashboard Views

Reporting Views

Export Views

Analytics Views

19.21 Naming Standards

Every table shall use singular names.

Examples

Employee

Contract

PayrollRun

PayrollTransaction

LeaveRequest

Notification

Columns use camelCase.

Primary Keys use UUID.

Foreign Keys reference UUID.

19.22 General Columns

Every transactional table contains:

id

createdAt

createdBy

updatedAt

updatedBy

version

status

isDeleted

deletedAt

deletedBy

These fields are mandatory unless explicitly excluded.

19.23 Relationships

Relationships shall be implemented using Foreign Keys.

One-to-Many

Many-to-One

Many-to-Many

Many-to-Many relationships require bridge tables.

19.24 Index Strategy

Every schema shall define indexes for:

UUID

Employee Number

Payroll Period

Company

Project

Status

Dates

Foreign Keys

Composite indexes shall be optimized according to reporting requirements.

19.25 Scalability

The schema supports:

Unlimited Companies

Unlimited Projects

Unlimited Employees

Unlimited Payroll Runs

Unlimited Historical Data

Millions of Transactions

Future Modules

19.26 Responsibilities

The Database Schema is responsible for:

Logical Organization

Data Integrity

Performance

Maintainability

Scalability

Historical Accuracy

Future Expansion

The physical tables defined in the next chapter shall follow these standards without exception.

End of Chapter 19

Chapter 20

Physical Database Design

20.1 Introduction

This chapter defines the physical database objects of the Enterprise Workforce Management & Payroll System.

The objective is to provide a complete blueprint for database implementation.

Every table shall include:

- Purpose
- Primary Key
- Foreign Keys
- Columns
- Constraints
- Indexes

- Business Notes

All tables are designed for PostgreSQL.

Primary Keys use UUID.

Audit columns are mandatory unless explicitly excluded.

20.2 Employee Table

Purpose

Stores the master employee record.

Primary Key

employeeId (UUID)

Columns

Column	Type	Description
employeeId	UUID	Primary Key
employeeNo	VARCHAR(30)	Employee Number
firstName	VARCHAR(100)	First Name
middleName	VARCHAR(100)	Middle Name
lastName	VARCHAR(100)	Last Name
fullName	VARCHAR(250)	Full Name
genderId	UUID	Gender
nationalityId	UUID	Nationality
religionId	UUID	Religion

Column	Type	Description
maritalStatusId	UUID	Marital Status
dateOfBirth	DATE	Birth Date
hireDate	DATE	Hire Date
status	VARCHAR(20)	Employment Status

Audit Fields

createdAt

createdBy

updatedAt

updatedBy

version

isDeleted

Indexes

Employee Number

Full Name

Status

Nationality

Hire Date

20.3 Employee Bank

Purpose

Stores employee bank information.

Columns

bankAccountId

employeeId

bankId

branchName

accountNumber

iban

currencyId

effectiveFrom

effectiveTo

status

Business Rules

One employee may have multiple bank accounts.

Only one account may be active during a payroll period.

20.4 Employee Document

Stores all employee documents.

Examples

Passport

Visa

National ID

Medical Certificate

Driving License

Employment Contract

Fields

documentId

employeeId

documentTypeId

documentNumber

issueDate

expiryDate

filePath

status

20.5 Employee Dependent

Stores employee dependents.

Examples

Spouse

Child

Parent

Fields

dependentId

employeeId

relationshipType

name

dateOfBirth

gender

nationality

eligibleForInsurance

eligibleForAirTicket

20.6 Employment Contract

Purpose

Stores employment contracts.

Columns

contractId

employeeId

companyId

employmentType

contractType

effectiveFrom

effectiveTo

basicSalary

currencyId

payrollFrequency

rulePackageId

companyPolicyId

status

Indexes

Employee

Effective Date

Company

Status

20.7 Salary Package

Purpose

Stores recurring payroll components attached to the contract.

Columns

salaryPackageId

contractId

payrollComponentId

amount

currencyId

paymentFrequency

effectiveFrom

effectiveTo

status

One contract may contain unlimited payroll components.

20.8 Employee Assignment

Purpose

Stores employee organizational assignments.

Columns

assignmentId

employeeId

companyId

businessUnitId

projectId

departmentId

sectionId

shiftId

costCenterId

effectiveFrom

effectiveTo

status

Employees may have multiple historical assignments.

20.9 Calendar

Columns

calendarId

calendarYear

calendarName

countryId

companyId

status

20.10 Payroll Period

Columns

periodId

calendarId

periodCode

periodName

startDate

endDate

payrollStatus

isClosed

Business Rule

One Payroll Period belongs to one Calendar.

20.11 Shift

Columns

shiftId

shiftCode

shiftName

startTime

endTime

workingHours

declaredOtHours

maximumHours

nightShift

weeklyOff

status

20.12 Time Type

Columns

timeTypeId

timeTypeCode

timeTypeName

category

paid

requiresApproval

rulePackageId

status

Examples

Present

Absent

Annual Leave

Sick Leave

Holiday

Friday

Training

Business Trip

20.13 Payroll Component

Columns

payrollComponentId

componentCode

componentName

category

calculationMethod

currency

paymentFrequency

status

Examples

Basic

Housing

Transport

Food

Bonus

Loan

Commission

Schooling

Penalty

Medical

20.14 Cost Center

Columns

costCenterId

costCenterCode

description

companyId

status

20.15 Cost Code

Columns

costCodeId

costCenterId

projectId

costCode

description

status

Business Rule

Unlimited Cost Codes.

No fixed CC1…CC15 structure.

20.16 Business Rules

Every table shall include:

UUID Primary Key

Audit Fields

Soft Delete

Version Number

Status

Effective Dating (where applicable)

Foreign Key Constraints

Indexes

20.17 Naming Standards

Table Names

Singular

Examples

Employee

Contract

Shift

PayrollRun

LeaveRequest

Notification

Column Naming

camelCase

Primary Keys

Id

Foreign Keys

Referenced Entity + Id

20.18 Performance

Large tables shall be partitioned.

Examples

Timesheet

PayrollTransaction

AuditLog

NotificationQueue

Settlement

Partitioning Strategy

Payroll Period

or

Year

depending on table type.

20.19 Responsibilities

The Physical Database Design defines the physical implementation of every business entity.

Subsequent chapters will define the remaining transactional tables:

- Timesheet
- Payroll
- Leave
- Overtime
- Provision
- Financial Transactions
- Settlement
- Workflow
- Reporting

These tables will follow the standards defined in this chapter.

End of Chapter 20

Chapter 21

Database Implementation Blueprint

21.1 Objective

This chapter defines the logical database implementation blueprint.

The purpose is **NOT** to define every table manually.

Instead, this chapter defines the database modules that shall be generated automatically from this document.

The complete PostgreSQL schema shall strictly follow all business rules defined in Chapters 1 through 20.

No additional business logic may be introduced unless explicitly approved.

21.2 Database Modules

The database shall consist of the following domains.

Employee Domain

Purpose

Employee master information.

Required Tables

Employee

EmployeeAddress

EmployeeContact

EmployeeEmergencyContact

EmployeeDependent

EmployeeEducation

EmployeeExperience

EmployeeSkill

EmployeeDocument

EmployeeMedical

EmployeeBank

EmployeeAttachment

EmployeeHistory

Company Domain

Company

BusinessUnit

Department

Section

Location

OrganizationStructure

CompanyPolicy

CompanyHolidayGroup

Project Domain

Project

ProjectPhase

ProjectArea

ProjectSection

ProjectCostCenter

ProjectCostCode

ProjectAssignment

Contract Domain

Contract

SalaryPackage

ContractAllowance

ContractDeduction

ContractAmendment

ContractHistory

LaborLawAssignment

CompanyPolicyAssignment

Assignment Domain

Assignment

ProjectAssignment

DepartmentAssignment

SectionAssignment

ShiftAssignment

SupervisorAssignment

AssignmentHistory

Calendar Domain

Calendar

PayrollPeriod

Holiday

HolidayGroup

HolidayAssignment

CalendarVersion

PayrollCalendar

Shift Domain

Shift

ShiftRule

ShiftBreak

ShiftRotation

ShiftCalendar

ShiftHistory

ShiftAssignmentHistory

Time Type Domain

TimeType

TimeTypeRule

TimeTypePayrollMatrix

TimeTypeProvisionMatrix

TimeTypeValidation

Payroll Component Domain

PayrollComponent

PayrollComponentCategory

PayrollComponentRule

PayrollComponentHistory

PayrollMatrix

Timesheet Domain

TimesheetHeader

TimesheetDetail

WorkSegment

AttendanceEvent

AttendanceImport

AttendanceException

CostAllocation

DailySummary

TimesheetApproval

TimesheetHistory

Leave Domain

LeaveRequest

LeaveBalance

LeaveAccrual

LeaveApproval

LeaveAttachment

LeaveSettlement

LeaveHistory

LeaveType

Overtime Domain

OvertimeRequest

DeclaredOvertime

UndeclaredOvertime

HolidayOvertime

FridayOvertime

NightOvertime

OvertimeApproval

OvertimeHistory

Financial Transaction Domain

EmployeeFinancialTransaction

Loan

LoanInstallment

SalaryAdvance

Bonus

Commission

Schooling

Penalty

RecurringTransaction

ScheduledTransaction

FinancialAdjustment

Payroll Domain

PayrollRun

PayrollEmployee

PayrollTransaction

PayrollComponentTransaction

PayrollBankTransfer

PayrollJournal

PayrollAudit

Payslip

PayrollHistory

Provision Domain

ProvisionTransaction

EOSProvision

LeaveProvision

AirTicketProvision

ProvisionAdjustment

ProvisionReversal

ProvisionHistory

Final Settlement Domain

Settlement

SettlementDetail

SettlementRecovery

SettlementApproval

SettlementPayment

SettlementHistory

SettlementAttachment

Workflow Domain

WorkflowTemplate

WorkflowStep

Approval

ApprovalHistory

Delegation

Escalation

WorkflowLog

Notification Domain

Notification

NotificationQueue

EmailNotification

SMSNotification

PushNotification

Reminder

NotificationHistory

Security Domain

SystemUser

Role

Permission

RolePermission

UserRole

LoginSession

PasswordHistory

ApiToken

Audit Domain

AuditLog

EntityHistory

PayrollAudit

IntegrationAudit

LoginAudit

ChangeHistory

Integration Domain

ImportJob

ExportJob

IntegrationQueue

ApiRequest

ApiResponse

FileImport

FileExport

MappingDefinition

Reporting Domain

Materialized Views

Dashboard Views

Summary Tables

Export Views

Analytics Views

21.3 Database Standards

The generated PostgreSQL database shall follow the following standards:

Primary Keys

UUID

Foreign Keys

UUID References

Audit Columns

Mandatory

Soft Delete

Enabled

Versioning

Enabled

Effective Dating

Enabled

Indexes

Optimized

Partitioning

Large Transaction Tables

History

Immutable

21.4 Generation Rules

The generated schema shall:

- Normalize data to at least Third Normal Form (3NF), except where denormalization is intentionally introduced for performance.
- Use PostgreSQL best practices.
- Avoid duplicated information.
- Support historical records.
- Support unlimited companies.
- Support unlimited projects.
- Support unlimited employees.
- Support unlimited payroll periods.
- Support future expansion.

21.5 Expected Deliverables

Based on this document, the implementation phase shall automatically generate:

- PostgreSQL Database
- ER Diagram
- Foreign Keys
- Indexes
- Constraints
- Materialized Views
- Partitions
- Flyway Migration Scripts
- Spring Boot JPA Entities
- Repository Layer
- Liquibase/Flyway SQL
- Initial Seed Data

21.6 Final Instruction for AI Development

This Functional & Technical Design Document is the single source of truth for the implementation of the Enterprise Workforce Management & Payroll System.

The AI shall generate the complete PostgreSQL database using the specifications defined in Chapters 1 through 21.

The AI shall not introduce additional business logic.

The AI may optimize the physical implementation while preserving all functional requirements.

All generated database objects shall remain fully consistent with the business architecture defined in this document.

End of Chapter 21

Chapter 22

System Administration, Security & Platform Management

22.1 Introduction

The System Administration Module is responsible for managing the entire Workforce Management & Payroll System.

This module controls:

- System Configuration
- Security
- Users
- Roles
- Permissions
- Payroll Period Lifecycle
- Audit Trail
- Monitoring
- Master Settings

The objective is to ensure the system is secure, auditable, configurable and manageable for enterprise organizations.

22.2 System Parameters

The system shall provide a centralized configuration module.

Examples include:

Company Information

Default Currency

Language

Time Zone

Date Format

Time Format

Number Format

Fiscal Year

Payroll Calendar

Default Shift

Default Rule Package

Password Policy

Notification Settings

Email Configuration

SMS Configuration

File Storage Settings

Backup Configuration

API Configuration

Integration Endpoints

System Logo

Company Branding

Changing system parameters shall not require software deployment.

22.3 User Management

The system shall support unlimited users.

Each user shall contain:

User ID

Username

Password

Employee Reference (Optional)

Full Name

Email

Phone

Language

Time Zone

Status

Last Login

Password Expiry

Account Lock Status

MFA Status

22.4 Authentication

Supported authentication methods:

Username & Password

Microsoft Active Directory (Optional)

LDAP (Optional)

OAuth2

OpenID Connect

Single Sign-On (SSO)

JWT Authentication

Refresh Tokens

Multi-Factor Authentication (MFA)

Future authentication providers may be added.

22.5 Authorization

The system shall implement Role-Based Access Control (RBAC).

Permissions are assigned to Roles.

Users inherit permissions from Roles.

Permission Levels include:

No Access

Read

Create

Update

Delete

Approve

Export

Import

Administration

22.6 Roles

Examples

System Administrator

HR Administrator

Payroll Manager

Payroll Officer

Project Manager

Department Manager

Timekeeper

Finance Officer

Auditor

Employee Self Service

CEO

Custom Roles

Roles shall be configurable.

22.7 Data Security

Access may be restricted by:

Company

Business Unit

Project

Department

Section

Employee Category

Payroll Group

Country

For example:

A Project Manager can only access employees assigned to his project.

A Payroll Officer can only access payroll groups assigned to him.

22.8 Payroll Period Lifecycle

Each Payroll Period passes through the following statuses:

Draft

↓

Open

↓

Processing

↓

Calculated

↓

Validated

↓

Approved

↓

Locked

↓

Bank Transfer

↓

Closed

Only authorized users may change the status.

22.9 Payroll Lock

Once Payroll is Locked:

Timesheets cannot be modified.

Leave cannot be modified.

Salary cannot be modified.

Payroll Components cannot be modified.

Overtime cannot be modified.

Financial Transactions cannot be modified.

Settlement cannot be generated.

Payroll becomes read-only.

22.10 Payroll Reopen

Payroll Reopen requires a formal workflow.

Example

Payroll Officer

↓

Submit Reopen Request

↓

Reason

↓

Payroll Manager Approval

↓

Finance Approval

↓

Payroll Unlocked

↓

Correction

↓

Payroll Recalculated

↓

Payroll Locked Again

↓

Payroll Closed

Every reopen operation shall be audited.

22.11 Data Freeze

Closing a Payroll Period automatically freezes related business data.

Frozen Modules:

Timesheet

Leave

Overtime

Payroll Components

Salary Package

Assignments

Shift Assignment

Cost Allocation

Financial Transactions

Only Payroll Reopen may release frozen data.

22.12 Audit Trail

Every business transaction shall generate an immutable audit record.

Audit captures:

Action

User

Date

Time

Module

Table

Record ID

Old Value

New Value

Reason

IP Address

Browser

Device

Session ID

Rule Version

Approval Reference

Audit records shall never be deleted.

22.13 Change History

Historical changes shall be available for:

Employee

Contract

Salary

Payroll

Leave

Timesheet

Overtime

Provision

Settlement

System Configuration

The system shall allow comparison between versions.

22.14 Activity Log

Every user action shall be logged.

Examples:

Login

Logout

Create Record

Update Record

Delete Record

Approve

Reject

Import

Export

Report Generation

Password Change

Configuration Change

22.15 Session Management

The system shall support:

Session Timeout

Concurrent Session Control

Force Logout

Remember Me (Optional)

Active Session Monitoring

Device Registration

22.16 Monitoring

The system shall monitor:

Online Users

Background Jobs

Payroll Processing

Queue Status

Database Health

API Performance

Disk Usage

CPU Usage

Memory Usage

Email Queue

Notification Queue

Integration Status

Error Logs

22.17 Backup & Recovery

The platform shall support:

Automatic Daily Backup

Incremental Backup

Point-in-Time Recovery

Database Replication

Backup Encryption

Restore Testing

File Backup

22.18 Security Policies

Password Rules:

Minimum Length

Complex Password

Password Expiration

Password History

Failed Login Attempts

Automatic Account Lock

MFA Enforcement

Idle Session Timeout

22.19 Compliance

The system shall comply with:

Qatar Labor Law

GDPR Principles (where applicable)

Company Security Policies

Internal Audit Requirements

Financial Audit Requirements

Data Retention Policies

22.20 Responsibilities

The System Administration Module is responsible for:

System Configuration

User Management

Security

Authentication

Authorization

Payroll Period Management

Payroll Lock

Payroll Reopen

Audit Trail

Activity Monitoring

System Monitoring

Backup

Recovery

Compliance

This module governs the operation, security and lifecycle management of the entire platform.

End of Chapter 22